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Space–time analysis of entrepreneurial ecosystems

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Abstract
Entrepreneurial ecosystems (EEs) are viewed as the combination or interaction of elements producing shared innovative and cultural values that support entrepreneurial activities within a territory. The literature has mainly taken a static view of the concept and has analyzed these ecosystems at national or at NUTS-2 regional level. This paper proposes the NUTS-3 provincial level as a spatial unit for measuring EEs in a dynamic perspective and focuses on the Italian case using data for the 2000–2018 period. The choice of NUTS-3 provinces is supported by theoretical arguments and it is confirmed by empirical results, which show some degree of heterogeneity in the level of EEs across provinces and some spatial patterns that would have been ignored using larger spatial units. The temporal dimension also reveals some intra-distribution mobility in the cross-sectional values of the provincial EE index. The paper contributes to advancing the research on EEs, has implications for policymakers and practitioners, and suggests directions for future studies.

Keywords Entrepreneurial ecosystem · Spatial analysis · Distribution dynamics · Italian provinces

JEL Classification L26 · R11 · O47

1 Introduction

In recent years the concept of entrepreneurial ecosystem (EE) has received increasing attention by researchers and policy makers, as demonstrated by the growing number of scientific articles, books, and literature reviews (see, for instance, De Brito & Leitão, 2021; Cao & Shi, 2020; Cavallo et al., 2019; Malecki, 2018; Wurth et al., 2021, among others). Stam, (2015, p.5) defines an EE a "set of interdependent actors and factors coordinated in such a way that they enable productive entrepreneurship within a particular territory". The EE concept emerged as a framework to better understand the nature of places in which entrepreneurial activity and innovation prospers, however it still lacks a shared analytical basis to clearly identify its definition, determinants, and effects. Several issues remain

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controversial (Alvedalen & Boschma, 2017). For instance, part of the literature criticizes the inconsistent use of the term ecosystem and its vague definition that adds no additional value to the scholarly discourse, to existing concepts like cluster, regional innovation system, triple-helix initiatives or public–private partnerships (Brown & Mason, 2017; Stam & Spigel, 2018). Others argue that the concept is under-theorized and conceptually fragmented (Alvedalen & Boschma, 2017). Still others that it is not always clear in what way proposed elements are connected in an EE (Stam, 2015), or which interactions matter most (Ács et al., 2014). The specific mechanisms through which ecosystems influence entrepreneurs are also not well-defined (Roundy & Fayard, 2019).

Despite the large amount of research devoted to investigating these issues, one still unaddressed aspect is a closer examination of the spatial content as well as the dynamic relationships that characterize an EE (Spigel, 2017). Researchers have criticized the EE literature because it proposed a static point of view that describes relationships within an EE without considering its evolution over time (Alvedalen & Boschma, 2017). Moreover, it is not clear which level of analysis the literature is targeting, e.g., city, region, or country (Stam, 2015). In a recent analysis, Lange and Schmidt (2020) argued that EE should be viewed as a bridging concept that not only opens a communicative trans-disciplinary exchange but also allows to integrate novel dynamic temporal and spatial elements to otherwise rather strict spatial boundaries of the concept.

We believe that these gaps should be addressed to improve our understanding of EEs, given that most of the factors that characterize EEs originate at local level and evolve over time. In other words, the paper answers the following questions: how does the dynamic analysis and increased granularity of measurement improve our understanding of EEs? How relevant are local factors for understanding EEs and their evolution? To answer these questions the paper (1) proposes a rigorous spatial unit for analyzing EEs, (2) measures EEs at this spatial scale level and in a dynamic perspective, (3) analyzes how EEs and the conditions that make up the EEs evolve over time and across spatial units, and then (4) assesses the sensitivity of the EE index to changes in the scale of the spatial unit, i.e., whether it is more appropriate to measure and examine the EEs at NUTS-3 provincial level or NUTS-2 regional level.[1]

In the paper, we adopt the following strategy. Regarding the unit of analysis, we propose the NUTS-3 provincial level, which we believe is the most appropriate unit to identify and measure EEs. To measure EEs, we follow the framework proposed by Stam (2015) and construct an index of EEs by modifying the version of Iacobucci and Perugini (2020a, 2020b), adjusted to account for the temporal dimension of the data, which goes from 2000 to 2018. We use Stam's framework not for theoretical reasons, but mainly because it allows to overcome the problem of data availability at this spatial scale while preserving a way, out of other possible, to conceptualize and theorize EE. We also examine the dynamics and the spatial dimension of the EE index and its components by employing the mean and the standard deviation, while to look at

[1] NUTS (Nomenclature of Territorial Units for Statistics) is the geocode standard for referencing the subdivision of countries for statistical purposes, adopted by the EU. The NUTS hierarchical classification divides territories into three levels, where the NUTS 1 level identifies larger socio-economic regions, the NUTS 2 level indicates basic regions, often considered in the definition of regional policies, and the NUTS 3 level identifies the smaller regions, which can be considered for specific analyses (https://ec.europa.eu/eurostat/web/nuts/background). In the rest of the paper, we refer to NUTS-1 as "geographical macro-areas", NUTS-2 as "regions", and NUTS-3 as "provinces". Moreover, we use the generic term "local" to define a small area, a limited place, a place generally within a city or a province.

its spatial pattern and to assess the heterogeneity across provinces and over time we compute the Theil index (Theil, 1971). Moreover, we apply the distribution dynamic approach developed by Quah (1993, 1996, 1997) in order to assess the extent of how far the provinces moved along the cross-sectional distribution of the EE index over time. Finally, for the analysis of the spatial correlation of the EE index and to assess the sensitivity of the EE index to changes in the scale of the spatial unit, we compute Moran's I statistic (Moran, 1948) and rely on regression analysis.

The paper makes several contributions to advancing the research on EEs. First, the paper provides a measure for assessing how the conditions that shape EEs develop. This is a relevant issue given the heterogeneity and complexity of EEs, thus allowing comparability.

The paper also propose the NUTS-3 provincial level as unit of analysis. This is a novelty in the EE literature. Indeed, researchers have considered the lack of studies at a smaller scale unit than the national level as a weakness in the EE literature, given that most of the factors that characterize EEs, vary largely between regions, provinces and, in some cases, cities. Providing a measure of EE at a low level of spatial disaggregation is extremely relevant given that most of the factors that characterize EEs, such as regulations, institutions and norms, infrastructure, city amenities, access to finance, demand, etc., vary largely between regions, provinces, and in some cases cities. Moreover, as the literature suggest, the nature of entrepreneurship is mainly local (Autio et al., 2014; Bosma & Sternberg, 2014; Cunningham et al., 2019; Fritsch, 2013; Sternberg et al., 2019; Tsvetkova, 2015).

The paper tells us more about the dynamics of the EEs, how they evolve, and how the relative strength of components changes, all issues that are less explored in the literature. The paper also provides further insights into whether the dynamics and evolution of EEs in one location is influenced by the EE endowment and development in neighbouring locations. Combining the spatial with the time dimension is necessary because as EE evolves in both directions (i.e., time and space), the importance and relative strength of components changes, and so do the policies necessary to sustain these systems over time (Alvedalen & Boschma, 2017; D. B. Audretsch et al., 2019; Brown & Mason, 2017; Mack & Mayer, 2016; Spigel, 2017).

The analysis carried out for the Italian case is also a novel contribution to the theoretical and empirical literature. This is particularly interesting given that Italy is characterized by a persistent economic dualism, which has historically resulted in a considerable disparity among the Northern and Southern regions and provinces in terms of economic development (Toniolo, 2013).

Finally, the paper contributes to the policy-making decision process. Indeed, empirical findings reveal that EEs are spatially heterogeneous and therefore targeted interventions are essential to foster and stimulate the conditions that shape the EE and entrepreneurial activities at sub-national level. Moreover, the EE literature has recognized that the causal mechanisms and actors involved in entrepreneurship activities occur at different spatial levels of analysis (D. B. Audretsch et al., 2019). Thus, findings may provide useful information for coordinating policy interventions at provincial, regional, and national level.

The paper is organized in the following sections. Section 2 provides a discussion of the literature that deals with the geographical and dynamic aspects of EEs. Section 3 defines the EE index, its measurement and the methodology used. Section 4 provides the empirical results. Section 5 discusses the implications while Sect. 6 describes the limitations of the paper. Finally, Sect. 7 draws the main conclusions.

2 The geography and the dynamics of EEs

In this section we highlight the most recent advancement in the debate regarding the spatial dimension of EEs and what the recent literature considers the main difficulties for defining a proper spatial boundaries. We also addressed why the recent literature on EEs regard the time dimension a relevant issue for better understanding EEs.

2.1 The spatial dimension

Apart from single case studies and some analysis of emerging ecosystems at city level (Cohen, 2006; Mack & Mayer, 2016; Motoyama & Knowlton, 2017; Pugh et al., 2019; Spigel, 2017), EEs are usually measured at national level (see, for instance, Ács et al., 2014; Isenberg, 2011; Stangler J. Bell-Masterson 2015). More recently, there have been some attempts to measure EEs at NUTS-1 (REDI), NUTS-2 regional (Guzman & Stern, 2020; Stam, 2015; Sternberg et al., 2019) or provincial level (Stam, 2018).

Researchers considered the lack of studies at smaller scale unit than the national level as a weakness of the literature on EEs, given that EE is a spatially bounded entity that follows a strong spatial logic. The potential and the need for a geographical dimension of EE at provincial or regional level is extremely important in those contexts where large territorial differences emerge in terms of start-ups and business growth, but also in terms of socioeconomic, organizational, and technological endowment, as is the Italian context, which is analyzed in this paper. Indeed, as we will see in the rest of the paper, most of the EE dimensions are not evenly distributed across space, so EE may vary not only between Northern and Southern Italian regions, as commonly expected, but also between adjacent provinces.

However, defining the spatial boundary of an EE is a difficult task for several reasons. First of all, the outputs and outcomes of EEs are the result of a complex set of actors and factors such as human capital, networks and institutions that occur in a temporal and varying regional setting (Leendertse et al., 2020). For instance, social networks play a crucial role for entrepreneurs to get access to resources like knowledge, finance and human capital (Stuart & Sorenson, 2005). This means that the spatial context could be a city, a region or a country, or it can also be other systems less strictly defined in space that create opportunities for firm creation and growth (Stam and van de Ven 2021). Moreover, some of the elements that make up the EE, such as finance or the quality of human capital, can be defined at a regional or provincial level, while others, such as regulations or investment in physical infrastructure, can be designed on both regional and national levels (D. B. Audretsch, Heger, et al., 2015; Audretsch, Link, et al., 2015).

One criterium to define the spatial boundary of the EE is to delimit the surrounding area where firms and individuals operate and interact. For instance, for entrepreneurs it could be the Local Labour Systems (LLS), i.e., geographical units delimited based on daily commuting of people for work reasons (Sforzi, 2009), or the same municipality when entrepreneurs interact with local banks. It is also possible to delimit the spatial context depending on the role assumed by the actors who operate within the EE. For civil servants it will be a particular administrative region, province, or municipality, for human capital a municipality or province where the university is established, for entrepreneurs it may be a municipality, a region, or even a country (Stam and van de Ven 2021).

Secondly, the definition of the spatial context is challenged by the presence of a multilevel context where formal and informal institutions interact at local (i.e., provincial in the Italian case), regional, and national level (Bosma et al., 2018). For this reason, some authors believe that although some level of spatial unit needs to be chosen for research purposes, multi-level methods are often the most appropriate ones for explaining entrepreneurship when considering sub-national spatial units (Hundt & Sternberg, 2016).

Thirdly, there is also the fact that although entrepreneurship is a local event, distant resources can be critical especially for new firms that are "born global" (Mahroum, 2016), but equally important is the firms' ability to integrate global knowledge and become successful in their home EE (Malecki, 2011). At the same time, multinational corporations are perceived to attract skilled employees to that territory, upgrade managerial skills of local firms, operate as a source for spin-off firms and provide business opportunities for firms to access global markets (Mason C.& Brown 2014). Moreover, digitalization and globalization increase entrepreneurs' opportunities to be exploited outside a cluster, causing entrepreneurship to lose its local dimension (Autio et al., 2018).

Finally, the choice of the spatial level of EE is constrained by data availability. Indeed, "there is often a disconnect between the theoretical definition of a region as integrated contiguous space and the political and census geography for which data are readily available" (Feldman & Lowe, 2015, p.1785). This limitation has so far constrained researchers to use data at national level or to conduct case studies analysis of specific high-tech places like the Boulder County and Phoenix in the US, Waterloo, Calgary and Victoria in Canada, and Lancaster in the UK (Cohen, 2006; Mack & Mayer, 2016; Pugh et al., 2019; Spigel, 2017).

2.2 The temporal dimension

The theoretical and empirical literature have also addressed the need for integrating the temporal dimension in regional studies of EEs (Alvedalen & Boschma, 2017; D. B. Audretsch et al., 2019; Brown & Mason, 2017; Mack & Mayer, 2016; Spigel, 2017). This perspective is necessary because EEs are complex and dynamic systems that evolve in space and over time. The EE components can also change and so do the institutions and the policies that are necessary to support these systems. A static analysis tends to miss the importance of institutional changes (Lowe & Feldman, 2017), and the type of support firms utilize through their own growth stages, from start-up to survival, growth, and exit (O'Connor & Reed, 2015).

To better understand the interdependencies between EE components and their evolutionary dynamics, scholars have proposed an industry lifecycle perspective, where the inherent dynamics lead to the birth, growth, maturity, decline, and re-emergence of EEs (Cantner et al., 2021; Mack & Mayer, 2016), or they have looked, based on a specific city's empirical setting, on how EEs are shaped by different governance designs, which create network interactions and relationships with local actors to support entrepreneurship (Colombelli et al., 2019). Other studies have also focused on case studies to shed light on the micro mechanism underlying the EE formation and evolution over time. This is the case of Ryan et al. (2020) who examined how multinational enterprises in the Irish city of Galway created and developed an EE. These studies go in the right direction of uncovering the relationship and the mechanisms that enable the formation of new ventures in a dynamic perspective, but they are confined to case studies. By contrast, there is a lack of research that provides quantitative empirical evidence on the complex dynamics of EEs. Also, in order

to capture the diversity and evolution of EEs more recent studies have started to use big-data approaches (Schwab & Zhang, 2019; von Bloh et al., 2020).

Following the above arguments, it is therefore important to assess how EE and its components change over time because it reveals how the dynamics of people, processes, and places encourage entrepreneurial behaviour, but also why some EEs fail to develop while others prosper. The time dynamic perspective is also relevant because it can provide useful insight about the cross-sectional distribution of the EE index, if for instance the EE levels in poorer (with a low endowment of the EE index) provinces catching up in relative terms with the those in the richer ones (with a high EE index endowment), but also if geographical proximity plays a role in explaining why EEs in some provinces are more dynamic than in others. Such analyses would help policy-makers to implement proper and adequate interventions.

3 Measuring the entrepreneurial ecosystem

Taking into account the research gaps opened at theoretical and empirical level, in the rest of the paper we aim to answer the following research question: how can EEs be measured at more granular level of analysis? To this purpose, in this section we first propose (1) a measure of EE, then (2) we examine the dynamics and spatial dimensions of the EE index and its components; (3) we explain the methodology used to assess to what extent provinces have moved along the cross-sectional distribution of the EE index over time; (4) and we assess the extent to which the conditions that make up an EE are geographically located, whether these are spatially correlated and how these are affected by changing the territorial unit of analysis.

3.1 Data

The composite index used in this paper to measure EE is made up of 11 variables/components. Variable selection followed the framework proposed by Stam (2018) to measure EE at the Dutch provincial level, but we also take into account how EE is measured in other studies that used data at national level (see, for instance, Ács et al., 2014; Isenberg, 2011; Stangler J. Bell-Masterson 2015). Despite the scarcity of data at NUTS-3 provincial level, we select variables that embody those domains and features characterizing an EE, as suggested by this literature. The 11 variables selected are the following: (1) Social capital; (2) Entrepreneurship Culture; (3) Physical Infrastructure; (4) Value Added; (5) Finance; (6) New knowledge; (7) Business services; (8) Firm survival; (9) Human capital; (10) Migration flow; (11) Population structure. The index is a modified version of Iacobucci and Perugini (2020a, 2020b), adjusted to account for the temporal dimension of the data. Table 1 provides details on the variables used and on the source of the data, while a detailed description of the selected variables is in the Appendix.

To ensure comparability and given that the variables have different measurement units, data for these variables have been normalized on a 0–10 scale using the min–max method, with higher values representing better performance.[2] In the second step, the 0–10 variable

[2] The min–max method normalizes indicators to have an identical range [0, 10] by subtracting the minimum value and dividing by the range of the indicator values (Nardo et al., 2005).

Table 1 The components of the Entrepreneurial Ecosystem (EE) index

	Variables/Conditions	Definition	Source	Year
1	Social capital	Measuring social networks, civic norms and trust: a) number of employees in social cooperatives over total employees; b) percentage of municipal waste object of separate collection on total municipal waste; c) number of protests on ordinary bills, drafts and bank checks over the resident population	ISTAT	2000–2018
2	Entrepreneurship culture	New firms registered per thousands of active populations	Movimprese	2000–2018
3	Physical Infrastructure	Motorway and railway potential accessibility index	Messina (2007)	2007
4	Value Added	Per-capita real Value Added	ISTAT	2000–2018
5	Finance	Amount of loans to households and firms as a share of Value Added	Bank of Italy and ISTAT	2000–2018
6	New knowledge	Capital stock of R&D by public and private firms as a equal share of Value Added, number of graduated students and number of firms in high-tech industries	ISTAT, Bronzini and Piselli (2009) and authors elaboration	2000–2018
7	Business services	Percentage of business service firms (M–N sectors in the Nace Rev.2; k74, k73, k71, K72, I63 in the Nace Rev.1.1) in the business population (G-O; G-K L)	Movimprese	2000–2018
8	Firm survival	Firm survival rate after 3 years	Authors elaboration on ISTAT data	2000–2018
9	Human capital	Estimation of a Mincer (1970) equation with years of schooling and work experience	Authors elaboration on ISTAT data	2000–2018
10	Migration flow	Net migration flows (immigrants less emigrants) over resident population	ISTAT	2000–2018
11	Population structure	Share of population in the 24 to 39 years old age group	ISTAT	2000–2018

scores are averaged using equal weight to obtain the final EE index. For more comparability of provincial rankings, the final scores for the EE index and its components are again normalized with the min–max method so that the range of the EE index and its components varies between 0 and 10. It should be noted that the min–max method could widen the range of values lying within a small interval, i.e. extreme values could distort the transformed component, but at the same time this method has the advantage of "rewarding" exceptional behaviour as it preserves the effects of the minimum and the maximum values of the distribution (Nardo et al., 2005). The not-normalized version of the final EE index provides similar results. Most importantly, our final EE index is robust to different weighting procedures, normalization, and aggregation methods. These robustness checks, which confirm the strength of the EE index, are explained in the Appendix in more detail.

Regarding the spatial level of analysis, our EE index is measured at the Italian NUTS-3 provincial level, the most disaggregated territorial level in relation to data availability. Indeed, data at the local administrative unit level (LAU or NUTS-4) are not available, as are data for the Local Labour Systems (LLS), which are geographical units between the LAU and the NUTS-3 that are delimited on the basis of daily commuting of people for work reasons.[3] As argued in the previous sections, we think that NUTS-3 is the more/most appropriate geographical level to capture the characteristics and the role of an EE and to account for the degree of spatial heterogeneity in the variables that compose the EE index. This is especially true for the Italian case where, as we will see in the next section, the conditions shaping the EE varies across adjacent territories. The data used for the analysis cover the 2000–2018 period.

3.2 Methodology

3.2.1 The temporal and spatial dimension of the EE

Based on the arguments provided in the previous sections, we use two statistical measures to examine the dynamics and the spatial dimension of the EE index and its components (point 1), that is the mean and the standard deviation. We also look at its spatial pattern and, to better assess the heterogeneity in the EE index at provincial level and over time, we compute the Theil index (Theil, 1971), a well-known measure of inequality that can be decomposed into between and within-group inequalities. In our case, groups are associated to NUTS-2 regions. The analysis is carried out for the EE index and for its components.

The visual inspection of the geographical distribution of the EE index or the synthetic measures of inequality and dispersion provide only a partial understanding of the evolution of the EE index over time and across space. This is because, persistent inequality in EEs across provinces is compatible with unchanging dispersion in the distribution of the EE index where, for instance, rich provinces (those with a higher endowment of the EE index) stay rich and the poorer provinces (with a low endowment of the EE index) stay poor, but it can also be consistent with marked changes in intra-distribution of individual provinces where the poor becomes rich, and the rich becomes poor. Intermediate cases, such as polarization or club-convergence (with two or more groups of provinces) in the distribution of the EE index can also be unnoticed through the empirical analysis of the Theil

[3] In 2011, the National Institute of Statistics classified 611 LLS with an average population of 97 thousand (ISTAT 2014).

index or other statistical measures of dispersion, such as the sigma convergence measure used in classical growth convergence analyses (Barro & Sala-I-Martin, 1992). Indeed, the studies that rely on the sigma (and the beta) convergence can only provide a summary of statistics for the evolution of distribution but not the details of mobility of provinces, and hence identical summary statistics can be derived from two distributions with entirely different mobility.

As a result, to assess the extent that provinces have moved along the cross-sectional distribution of the EE index over time (point 2) we apply the distribution dynamic approach developed by Quah (1993, 1996, 1997). This approach focuses the attention on the external shape of the distribution and on the movement within the distribution of the EE index over time (Fingleton & López-Bazo, 2003). As for external shape, it could be that over time the distribution of the EE index collapses to a single point, implying a convergence process by poorer provinces approaching the level of EE index of the rich ones. Or the external shape of the distribution might indicate an increase of dispersion of the EE index over time. The external shape might also remain unaltered (e.g., poor provinces become rich and vice-versa), thus the focus on intra-distribution mobility might reveal how provinces transit from any point in the distribution to any other point at a future time (Calcagnini & Perugini, 2019b).

The distribution dynamic approach requires estimating the probability of a province to staying poor or becoming rich. An estimate of the external shape typically involves nonparametric estimation of the density function via the kernel method (Silverman, 1986), while intra-distribution dynamics can be analyzed through the estimation of a stochastic kernel (Stokey et al., 1989) for the distribution of EE index over the period under analysis. The stochastic kernel is merely the counterpart of the transition probability matrix where the number of states (the values taken by the EE index) tends to infinity. The stochastic kernel can be drawn in a two dimension contour plot with the probability density function of EE at time $t+s$ on the x-axis and at time t on the y-axis. It can also be drawn in the corresponding three dimensional plot with the conditional density estimated, i.e., the stochastic kernel, on the vertical axes for each pair of points in the t and $t+s$ space. In the three dimensional plot for each value of EE at time t it shows the probability density s periods ahead, conditional on the density of that value at t. That is, it provides the probability of a province starting with any value of the EE index at t ending up at any of the values after s years.[4]

When most of the probability density function, as represented by the two dimensional contour plot, is concentrated along the diagonal, there is a lack of mobility in the variable of interest. In contrast, a tendency towards convergence would be signaled by a counterclockwise movement around the center of the 45-degree line. Complete convergence is represented by a concentration of most of the graph around the vertical 1-value of the time $t+s$ axis and parallel to the time t axis (the 1-value represent the average level of the EE index). Finally, the formation of two (or more) separate modes is a signal of a tendency toward polarization.[5]

The stochastic kernel can also be used to describe movements between two distributions, i.e., an actual unconditional distribution and a conditional distribution (Quah, 1996).

[4] Lines in the two dimensional plot connect points at the same height on the three-dimensional plot, that is, points with the same density.

[5] An analytical description of the nonparametric estimation of the density function and the stochastic kernel is in the Appendix.

The latter is the distribution that prevails when other variables affect future values of the EE index. In this case, when the mass of probability lies along the main diagonal the conditioning factor is not responsible for the variation in the actual distribution. In contrast, when a factor is responsible for most of the dispersion, conditioning out its effect will result in a kernel parallel to the unconditional distribution (so that provinces share similar values of the EE index when the effect of the factor is removed, regardless of their position in the original distribution).

To determine if geographical factors may explain provincial EE intra-distribution dynamics, it is possible to compare the initial unconditional EE distribution with the end of period space conditioned distribution. To this purpose, it is necessary to pre-filter the data to take out the influence of the conditioning variable before the analysis. In our case we are interested to see if geographical proximity influences the distribution of the EE index. Therefore, to determine the space conditioned distribution, we compute the EE of a province relative to the average of its geographical neighbours. As we have seen above, there are two distinct clusters of provinces, one in the North of Italy, where rich provinces tend to be close to other rich provinces, and one in the South of Italy, where poor provinces tend to be close to other poor provinces. Therefore, it is interesting to see how the spatial conditioning factor influence the distribution and the transition dynamics of the EE index. The conditional distribution can be interpreted as the part unexplained by geographical proximity. If no differences exist between the unconditional and the conditional distribution, then one may conclude that geographical proximity does not matter for the transition dynamics of the EE index.

3.3 Robustness check: spatial and regression analysis

As for the analysis of the spatial correlation of the EE index and the choice of the spatial unit of analysis (point 3), we compute Moran's I statistic (Moran, 1948) and rely on regression analysis. Moran's I compares the value of the observed variable at any location with the value of the same variable at neighboring locations.[6] The values of Moran's I statistic can also be plotted in a diagram, with the four different quadrants of the scatter plot identifying four types of local spatial associations between a province/region and its neighbours: going clockwise from top right we have HH (High-High), LH (Low–High) LL (Low-Low), and HL (High-Low). For instance, in the HH quadrant, the Moran scatter plot represents the high EE provinces/regions that are surrounded by high EE neighbours. Similar logic follows for the other quadrants. The spatial correlation in the EE index is measured at both the NUTS-3 provincial level and at the NUTS-2 regional level so we can compare to what extent the degree of spatial correlation varies when the scale and the shape of the spatial unit of analysis change.

In the literature the sensitivity of statistical results to the modification in the shape and the size of the geographical units is known as the Modifiable Areal Unit Problem (MAUP) (Fotheringham & Wong, 1991). It is argued that classical statistical techniques should be used with great caution when applied to data from spatial units because the statistical results might be unreliable and unstable depending on the definition of the spatial unit

[6] The Moran's I correlation coefficient range between − 1 and 1: the value 1 means perfect positive spatial autocorrelation (high values or low values cluster together), while -1 suggests perfect negative spatial autocorrelation, and'0' implies perfect spatial randomness. To quantify the connections between provinces a weighting matrix W is used. More details on the W selected is in footnote 16.

investigated (Fotheringham & Wong, 1991). This issue has been largely investigated in the literature on economic geography and on agglomeration economies (Arbia, 2012; Briant et al., 2010; Fotheringham & Wong, 1991; Openshaw, 1981; Wong, 2009), but the evidence on the effects of spatial aggregation and the choice of the spatial unit of analysis is mixed. However, despite a substantial literature on the potential influence of the MAUP, there is no empirical investigation of its effect on the entrepreneurial modeling practice.

To test the hypothesis that the appearance of EE clusters is dependent on the administrative scale level due to the MAUP, we also consider its incidence in the context of multivariate regression analysis. To carry out this exercise we estimate the following panel-data model:

$$EEBR_{it} = \alpha_0 + \alpha_1 EE_{it} + \alpha_k \sum_{K=1}^{N} X_{kit} + F_t + \varepsilon_{it} \quad (1)$$

where the dependent variable EEBR denotes the employer enterprise birth rate in province/region i at time t, EE denotes our entrepreneurial ecosystem index, and X_K are the k control variables, which include trade openness, a variable that measures industry heterogeneity (computed as the inverse of the Herfindahl–Hirschman index calculated on the number of active firms), and a measure of agglomeration economies, i.e. population.[7] Finally, F_t controls for time fixed effects, and ε_{it} is an error term.[8] Data refer to the 2000–2013 period.[9] The dependent variable, calculated as a ratio over the adult population, measures the capacity to start up a firm with employees (Ahmad & Hoffman, 2008).[10] For robustness, we also use the number of high growth enterprises as dependent variable (Coad et al., 2014).[11] Equation (1) follows previous studies, which investigated the impact of EEs on entrepreneurial activities in terms of creation of high-growth new firms (Mason C.& Brown 2014) or start-up rate of new firms (D. B. Audretsch & Belitski, 2017).

Given the panel structure of the data, we estimate Eq. (1) with random effects.[12] In order to investigate the generating process of possible spatial patterns in the EE index and to quantify the spatial dependency and spatial spillovers, we also considered spatial regression models that differ according to the way in which the error terms are specified: the spatial autoregressive model (SAR), and the spatial error model (SEM) (LeSage & Pace, 2009).[13] The spatial dependence in the SAR is incorporated into the model as the relationship between the dependent variable and the spatially lagged dependent

[7] The selection of the control variables, i.e., trade openness, industry heterogeneity, and population, that may have an impact on entrepreneurship follows previous literature (D. B. Audretsch and Vivarelli 1996; Scholman et al., 2015). Data description and descriptive statistics are in the Appendix.

[8] In 2018, the number of Italian NUTS-3 provinces is 110 and the number of NUTS-2 regions is 20. However, during the period 2000–2013, the number, and the political and geographical structure of Italian provinces changed. To keep the number of provinces constant over the period of analysis, we use data for 103 provinces and 20 regions.

[9] Data for the dependent variable are not available for more recent years.

[10] This indicator refers to the birth of an enterprise that has at least one employee in the birth year and of an enterprise that existed before the year in consideration but were below the threshold of one employee (OECD and of the European Communities 2008).

[11] This is defined as number of firms with average growth in the number of employees greater than 20 percent per year over a three-year period and with ten or more employees at the beginning of the period (OECD and of the European Communities 2008).

[12] The Hausmann test suggests the random effect is preferred to the fixed effect model, while the likelihood ratio test shows that the panel-level variance component in the random effects model is significant.

[13] Model selection was based on testing the constraint of the spatial parameters (Belotti et al., 2017).

variable(W*EEBR), where W is the spatial weighing matrix. The SEM comprises spatial autocorrelation with an error term and a spatially lagged error term ($W * \varepsilon_{it}$). A significant spatial autoregressive term (ρ for the SAR and λ for the SEM) indicates strong spatial dependence. Like the computation of Moran's I, Eq. (1) is estimated using data at both the NUTS-3 provincial level and at the NUTS-2 regional level. It should finally be noted that if$\rho \neq 0$, then the interpretation of the coefficient of the EE index parameter in the SAR is different from the conventional least square interpretation. This is because any change to the EE index or any other explanatory variable in a single province/region can affect the dependent variable in all provinces/regions. This is the result of the simultaneous spatial dependence model that considers other regions' EEBR, and these are determined by the characteristics of those regions. As a result, to interpret the coefficient of the SAR we need to compute the direct and indirect (spatial spillover) effect. The direct effect represents the expected average change across all observations for the dependent variable of in a particular province/region due to an increase of one unit for a specific explanatory variable in this region. The indirect effect represents the changes in the dependent variable of a particular province/region arising from a one unit increase in an explanatory variable in another region (LeSage & Pace, 2009).

4 Empirical results

4.1 Spatial and time dimensions

In Fig. 1 we plot the average value of the EE index and its standard deviation from 2000 to 2018. The tendency for both measures to increase until the start of the recent financial crisis in 2008 and then to gradually decline up to more recent years is can clearly be seen. This suggests that over the last two decades there has been a general deterioration of the conditions that make up the EE but also less dispersion in the value of EE index among provinces.

Looking at the single variable composing the EE index, the mean values of "Firm survival", and "Population structure" followed the same path of the EE index, while other variables such as "Business services" and "Entrepreneurship culture", registered an opposite trend, that is their values have been gradually increasing since 2008 (Table 2). A steady improvement of the mean values is observed for "Human capital", "New knowledge", "Migration flow", and to a lesser extent, "Social capital".

A deterioration of the mean values is instead observed for "Value Added" and "Finance", i.e., the EE components more directly influenced by the financial crisis. In terms of standard deviation, it is also interesting to note that since 2008 and in contrast to what has been observed for the overall EE index, the degree of dispersion increased for "Population structure", "New knowledge", and "Human capital". For "Value Added", and "Finance" instead we observed less heterogeneity among provincial values, with the standard deviation falling to lower levels in 2018. This might suggest a possible convergence process towards lower mean levels for these variables. Similar paths have been followed by "Social capital" and by "Firm survival". Indeed, since 2008 the degree of dispersion has reversed its trend and decreased. By contrast, from 2008 New knowledge, Human capital, and Population structure were the variables that contributed more to amplify the dispersion of the provincial EE index, which is measured by the standard deviation. Overall, since the financial crisis the EE index has deteriorated, but some of its components have improved.

Going into more detail at provincial level, Table 3 shows the values of the EE index of 15 provinces that are at the top and at the bottom of the rank in 2018 and in 2000 (the rank for all the provinces is in Table 7 in the Appendix). The top 15 provinces are all located in the North of Italy and the bottom 15 provinces are all located in the South, except for the province of Rieti which is in the Lazio region in the Centre of Italy. Data also reveal that the province of Milano is ranked at the top of the table in both years while the province of Rome is the province that improved the most. The last column in Table 3 shows the absolute change in the value of the EE index between 2018 and 2000. It is interesting to note that the values of the EE index for the top 15 provinces have on average remained unchanged while for the 15 provinces that are at the bottom of the rank there has been a slight improvement, by 0.2 points. In other words, over time the poorest provinces have approached the level of EE index of the richest ones.

A complete description of the EE index at provincial level is provided by its geographical representation in Fig. 2, where the spatial pattern of the EE index distribution is divided in deciles for some selected years.[14] In all maps the darker areas, associated with higher values of the index, are mainly located in the Northern provinces, suggesting that the conditions for entrepreneurship are more favorable within the more developed area of the country. This is also the case for all the EE components, except for Business services and for Population structure, which do not show any spatial pattern.[15]

Now we are going to investigate the time trends of the EE index and its components in greater detail, and we are going to assess if and to what extent the variability of the EE index is spatially concentrated. We begin by assessing the degree of provincial heterogeneity in the EE index through the Thiel index. In 2000, the Theil coefficient for the EE index at provincial level is 0.154 but this has gradually decreased over time, reaching 0.096 in 2018 (Fig. 3). However, while at the beginning of the period about 85% of the total inequality was explained by EE index disparity between regions, the within variability increased substantially especially since 2008: the percentage of within regional variability on the total Theil index was 15.7% in 2000 and 21.4% in 2018. Table 4 complements the results of the Theil index by showing the differences in the value of the EE index between the top and the bottom performer at provincial level that belong to the same NUTS-2 region. There are some marked differences, which are hidden by regional average. For instance, in the Lombardia region the province of Sondrio has an EE endowment largely lower than the more industrialized province of Milan. Similarly, for the Lazio region where the EE in the province of Rome is more developed than in the smaller provinces of Frosinone or Rieti. These large differences do not disappear over the years, suggesting that in some cases the connections between actors, institutions, and entrepreneurs are deeply rooted and well-functioning in some provinces but they do not contribute to improve the EE in other neighboring provinces.

Regarding the single variables that compose the EE index, data show a higher inequality in the "New knowledge", the "Migration flow" and, until 2010, the "Social capital" variable than in the EE index (Fig. 4), which partially confirm the indications provided above by the standard deviation. For these variables the within region variability in also large. For instance, on average the within variability for the "Social capital" accounts for around 60% of its total variability, a percentage four times higher

[14] We select the 2000, 2008, and 2018 years for reason of space. 2008 is selected as the year when the financial crisis started.

[15] The maps of all the components and of all the years are available from the authors upon request.

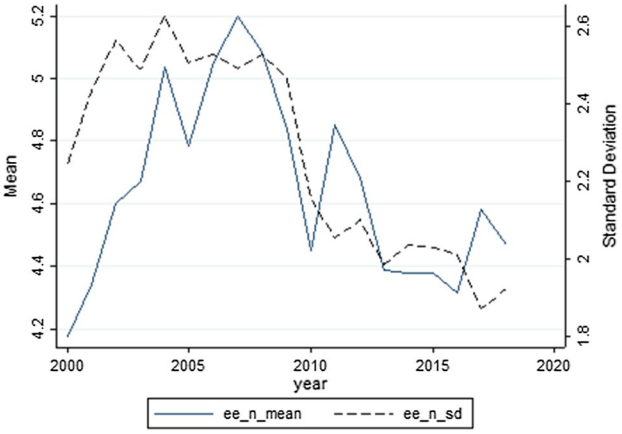


Fig. 1 Mean and standard deviation of the EE index (2000–2018)

Table 2 Summary statistics of EE index and its components in selected years

	Mean			Standard deviation		
	2000	2008	2018	2000	2008	2018
Entrepreneurial Ecosystem	4.2	5.2	4.5	2.2	2.5	1.9
Social capital	4.6	5.2	5.3	2.6	2.7	1.9
Entrepreneurship culture	4.1	2.8	3.5	2.1	1.7	1.7
Physical Infrastructure*	3.9	3.9	3.9	1.8	1.8	1.8
Value added	5.0	4.0	2.9	2.3	2.4	1.9
Finance	2.9	2.8	1.8	1.9	1.9	1.2
New Knowledge	2.3	2.5	3.6	2.3	2.1	2.4
Business services	2.9	2.7	3.5	1.6	1.6	1.7
Firm survival	5.1	5.5	4.9	1.3	1.9	1.6
Human capital	4.9	5.2	6.2	2.0	2.2	2.4
Migration flow	4.9	5.3	5.8	2.1	2.2	2.0
Population structure	4.6	8.0	4.8	2.6	1.3	2.2

*Values refer to 2007

than the within variability of the EE index (Fig. 5). Interestingly, throughout the period under observation most of the within regional inequality is accounted for by the same 4 regions, i.e., Lombardia, Lazio, Calabria, and Sicilia. In 2000, 8.5% of the total (15.7%) within region variability in the EE index are due to these regions. This percentage significantly increases from around 2008/2010 and reaches 13.1% in 2018 (Fig. 6).

Data also reveal that since 2010 these regions, apart from the Calabria region that is constantly at the bottom of the rank, experienced relatively large changes in the mean value of the EE index (Table 5). For instance, Lazio saw the largest increase while in Sicilia and, to a lesser extent, in Lombardia the mean values of the EE index decreased substantially. This suggests that in these regions the conditions that made up the EE changed considerably at the expenses of an increased heterogeneity within their territories, i.e., among their provinces. The last column of Table 5 also shows that between

Table 3 EE index of the top and bottom 15 provinces in 2018, in 2000, and absolute changes

Rank in 2018	NUTS–3 Provinces	NUTS–2 Regions	EE index in 2018	EE index in 2000	absolute change
1	Milano	Lombardia	10.0	10.0	0.0
2	Bologna	Emilia–Romagna	8.4	8.0	0.4
3	Roma	Lazio	7.9	6.3	1.6
4	Parma	Emilia–Romagna	7.8	8.4	–0.6
5	Torino	Piemonte	7.2	6.9	0.3
6	Padova	Veneto	7.2	7.1	0.1
7	Modena	Emilia–Romagna	7.1	7.1	0.0
8	Reggio nell'Emilia	Emilia–Romagna	7.1	8.5	–1.4
9	Bolzano/Bozen	Trentino–Alto Adige/Südtirol	6.9	6.4	0.5
10	Novara	Piemonte	6.8	6.9	–0.1
11	Forlì–Cesena	Emilia–Romagna	6.8	6.7	0.1
12	Cuneo	Piemonte	6.5	5.3	1.3
13	Rimini	Emilia–Romagna	6.5	7.3	–0.8
14	Brescia	Lombardia	6.5	7.4	–1.0
15	Verona	Veneto	6.4	6.5	–0.2
88	Oristano	Sardegna	2.1	1.4	0.7
89	Ragusa	Sicilia	2.1	2.0	0.0
90	Catanzaro	Calabria	2.1	2.3	–0.2
91	Rieti	Lazio	2.1	2.6	–0.6
92	Siracusa	Sicilia	2.0	1.9	0.1
93	Cosenza	Calabria	2.0	1.6	0.4
94	Messina	Sicilia	1.9	0.7	1.2
95	Taranto	Puglia	1.8	1.1	0.7
96	Palermo	Sicilia	1.7	1.2	0.5
97	Trapani	Sicilia	1.6	1.0	0.5
98	Vibo Valentia	Calabria	1.0	1.1	–0.1
99	Reggio di Calabria	Calabria	0.7	0.2	0.5

Table 3 (continued)

Rank in 2018	NUTS–3 Provinces	NUTS–2 Regions	EE index in 2018	EE index in 2000	absolute change
100	Caltanissetta	Sicilia	0.7	0.6	0.1
101	Enna	Sicilia	0.5	0.2	0.4
102	Agrigento	Sicilia	0.4	0.0	0.4
103	Crotone	Calabria	0.0	1.3	–1.3

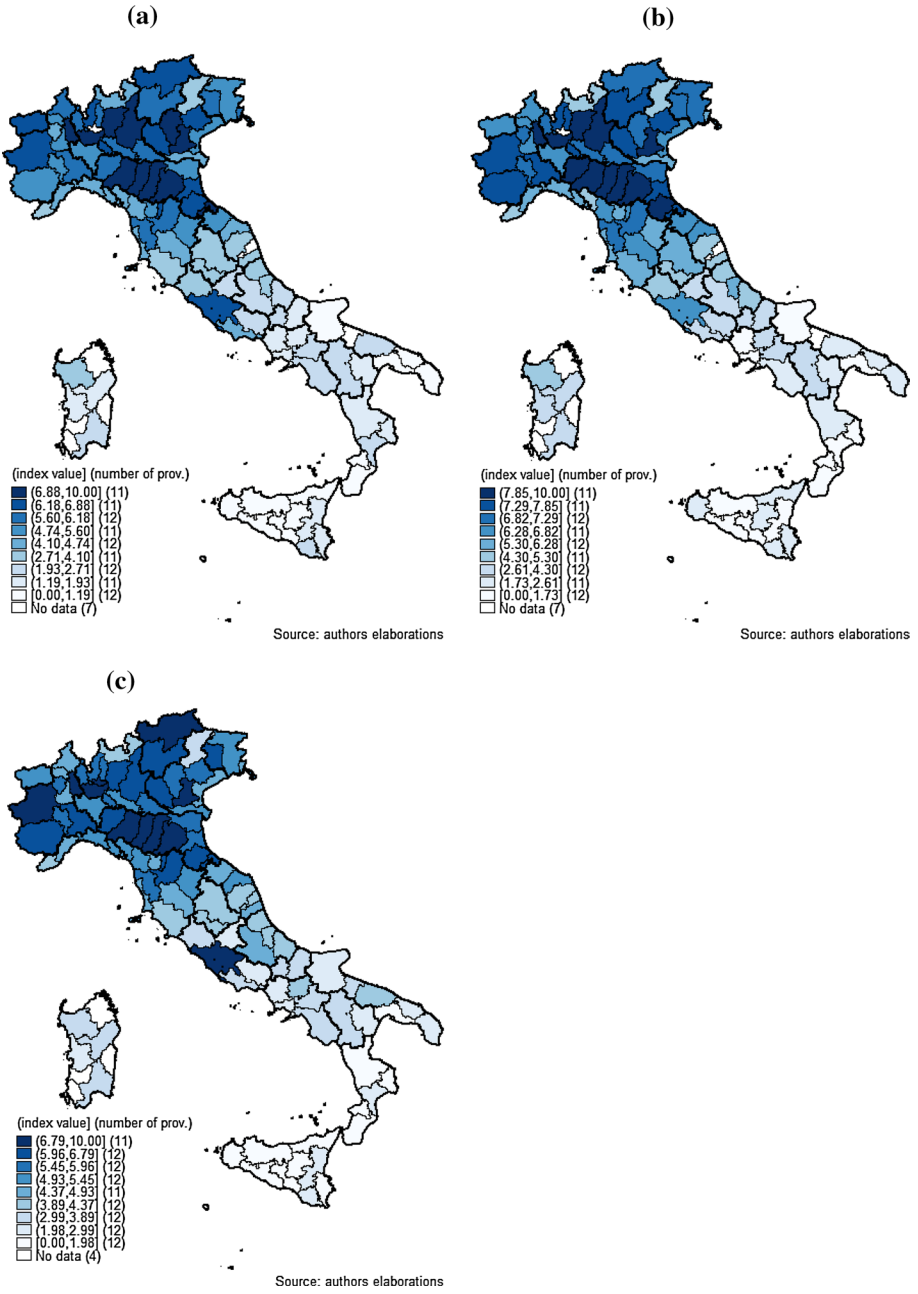


Fig. 2 Geographical distribution of the EE index at NUTS-3 provincial level in **a** 2000, **b** 2008, and **c** 2018. Black solid line delimits NUTS-2 regions. (darker areas mean higher values of the EE index)

2018 and 2000 the EE index improved more in the Southern regions than in the Northern-Centre regions. This is not surprising given what has been observed at provincial level.

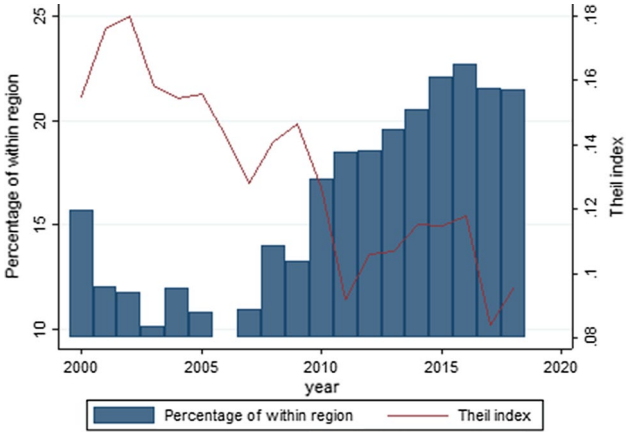


Fig. 3 Theil index (right axes) and the within regional variability (as a percentage of the total inequality) (left axis) (2000–2018)

However, apart from a few exceptions, the evidence provided so far and the maps in Fig. 2 confirm the presence of intra NUTS-2 regional variability in the values of provincial EE index (NUTS-2 regions are delimited by black solid lines in the maps in Fig. 2). These results support our efforts to analyze the EE at a more disaggregated territorial level than the regional or the national level. The above results also show that over time the value of the EE index of the poorest provinces moved toward the mean value of the richest ones. Indeed, if we estimate a growth regression model with EE data for all provinces, we obtain an absolute convergence rate, denoted by the beta-convergence, of about 2.8% (Mankiw et al., 1992). However, as mentioned in the Methodology section, a proper and complete evaluation of the convergence process requires the analysis of the transition dynamics of the distribution of EE index, given that studies of sigma and beta convergence cannot provide details of provincial mobility.

This is done using the distribution dynamic approach. The empirical results are shown in Figs. 7 and 8. The time evolution and changes in the external shape of the distributions over 2000–2018 are shown in Fig. 7a. More specifically, Fig. 7a shows the kernel estimates of the relative EE index densities for the initial and final years under analysis, i.e., 2000 and 2018, as well as for the ergodic distribution. While in the year 2000 the estimated density shows two well-separated peaks, the lowest at a value below the average (around the value of 0.50 times the Italian average EE index), and the other at around 1.30 times the Italian average EE index, in 2018 the density function collapses into a unimodal distribution, with the peak at around 1.10 times the Italian national average EE index. This is an indication of a convergence process in the EE index, with poor provinces becoming rich and rich becoming poor.

This tendency is more visible in the long run (the ergodic distribution), as a larger proportion of the probability mass is concentrated around a peak at about the same value of the 2018 EE index distribution. The contour plot in Fig. 7b and three dimensional plot in Fig. 7c, which provides information on the intra-distributional mobility of provinces over time, confirm the changes in the relative positions in the EE index distribution that emerged from the analysis of the external shape of the kernel densities. Indeed, Fig. 7b, c show that although a large portion of the probability mass is concentrated along the 45

Table 4 Top (first row) and bottom (second row) provinces within the same NUTS-2 region in selected years

	2000		2008		2018	
NUTS-2 regions	NUTS-3 provinces	EE	NUTS-3 provinces	EE	NUTS-3 provinces	EE
Piemonte	Novara	6.9	Novara	8.2	Torino	7.2
Piemonte	Biella	4.6	Vercelli	6.3	Verbano-Cusio-Ossola	4.5
Lombardia	Milano	10.0	Milano	10.0	Milano	10.0
Lombardia	Sondrio	4.2	Sondrio	5.3	Sondrio	4.3
Trentino-Alto Adige	Bolzano	6.4	Trento	7.9	Bolzano	6.9
Trentino-Alto Adige	Trento	6.1	Bolzano	6.8	Trento	6.2
Veneto	Padova	7.1	Padova	8.4	Padova	7.2
Veneto	Belluno	3.7	Belluno	4.7	Belluno	3.7
Friuli-Venezia Giulia	Pordenone	5.9	Pordenone	7.3	Pordenone	6.3
Friuli-Venezia Giulia	Trieste	5.2	Gorizia	5.1	Gorizia	4.5
Liguria	Genova	4.7	Genova	6.4	Genova	5.1
Liguria	Imperia	3.4	Imperia	5.1	Imperia	4.3
Emilia-Romagna	Reggio nell'Emilia	8.5	Bologna	9.6	Bologna	8.4
Emilia-Romagna	Ferrara	5.1	Ferrara	6.6	Ferrara	5.6
Toscana	Prato	6.4	Firenze	7.3	Firenze	6.1
Toscana	Grosseto	3.8	Massa-Carrara	4.9	Grosseto	4.0
Umbria	Perugia	4.1	Perugia	5.3	Perugia	4.2
Umbria	Terni	3.3	Terni	5.1	Terni	4.1
Marche	Pesaro e Urbino	4.8	Ancona	6.6	Ancona	5.1
Marche	Macerata	3.2	Macerata	4.8	Macerata	4.0
Lazio	Roma	6.3	Roma	6.6	Roma	7.9
Lazio	Frosinone	2.0	Frosinone	3.2	Rieti	2.1
Abruzzo	Pescara	3.7	Pescara	5.2	L'Aquila	4.4
Abruzzo	L'Aquila	2.0	L'Aquila	2.7	Teramo	4.0
Molise	Isernia	2.5	Campobasso	3.1	Isernia	3.3
Molise	Campobasso	1.5	Isernia	2.7	Campobasso	3.2
Campania	Salerno	2.3	Avellino	3.5	Benevento	4.1
Campania	Napoli	1.6	Napoli	0.9	Napoli	2.4
Puglia	Bari	2.0	Bari	2.6	Bari	4.0
Puglia	Foggia	0.5	Taranto	1.4	Taranto	1.8
Basilicata	Potenza	2.3	Potenza	2.8	Potenza	3.1
Basilicata	Matera	1.5	Matera	2.0	Matera	3.0
Calabria	Catanzaro	2.3	Cosenza	2.0	Catanzaro	2.1
Calabria	Reggio di Calabria	0.2	Reggio di Calabria	0.0	Crotone	0.0
Sicilia	Ragusa	2.0	Ragusa	2.2	Catania	2.3
Sicilia	Agrigento	0.0	Caltanissetta	0.1	Agrigento	0.4
Sardegna	Sassari	3.3	Cagliari	4.5	Cagliari	3.8
Sardegna	Oristano	1.4	Oristano	2.6	Oristano	2.1

*Regions are ordered from their geographical position, from North to South of Italy

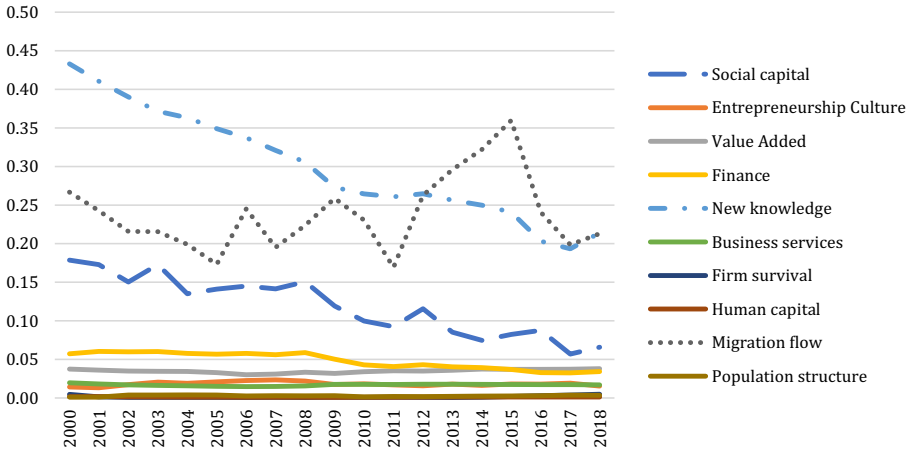


Fig. 4 Theil index by variables (2000–2018)*. *The "Physical Infrastructure "variable is not included in the Figure since data are available only for the year 2007

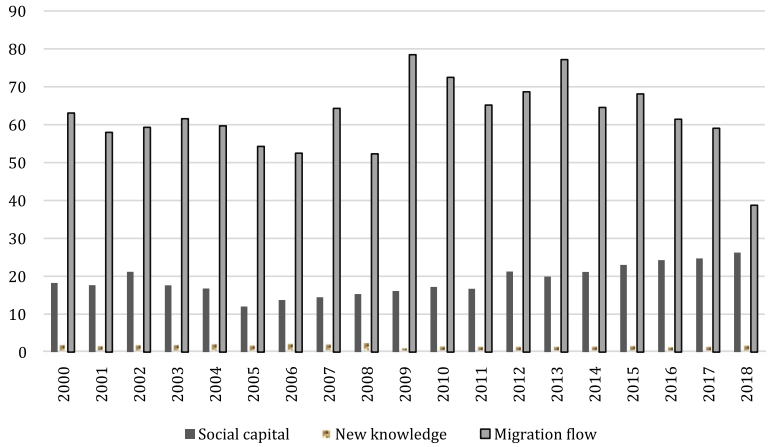


Fig. 5 Within regional variability (Theil index) for selected variables (2000–2018) (as a percentage of the total inequality)

degree diagonal, thus suggesting persistence, the two peaks respectively for rich provinces (clustered initially around 1.20 times average EE index), and for poor provinces (clustered around 0.30 times average EE index) lie above and below this line. The counter-clockwise rotation around the main diagonal line in two directions suggests that some of the poorest provinces tend to increase their relative EE endowment more likely over the considered period while the opposite is true for richer provinces.

Figure 8 is the counterpart of Fig. 7 and shows the distributions (a) and the stochastic kernel in terms of two (b) and three (c) dimensional plots with the space-conditional end of period EE index, i.e., data are pre-filtered on geographical proximity. Figure 8a shows that the external shapes of the 2018 and ergodic EE index distribution becomes unimodal but compared to Fig. 7a the probability densities are more concentrated around the (relatively lower) peak of 0.90 times the average EE index. This means that

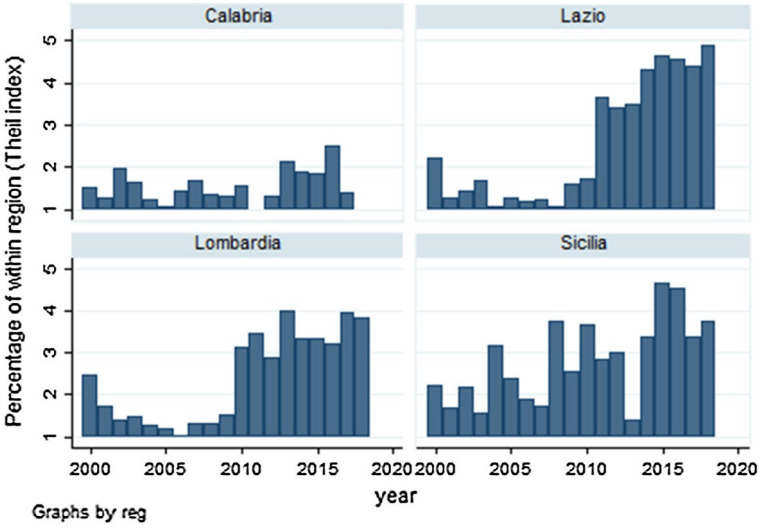


Fig. 6 Top 4 regions with largest percentage of within regional variability (Theil index)

Table 5 EE index at NUTS-2 regional level and absolute changes

NUTS–2 Regions	EE in 2018	EE in 2010	EE in 2000	Absolute changes (2018–2010)	Absolute changes (2018–2000)
Piemonte	8.4	8.6	7.8	–0.2	0.6
Valle d'Aosta/Vallée d'Aoste	7.7	7.2	7.3	0.4	0.3
Lombardia	9.7	10.0	10.0	–0.3	–0.3
Trentino–Alto Adige/Südtirol	10.0	9.0	8.1	1.0	1.9
Veneto	7.3	7.5	7.9	–0.2	–0.6
Friuli–Venezia Giulia	6.3	7.2	7.0	–0.9	–0.7
Liguria	6.1	6.4	5.5	–0.2	0.6
Emilia–Romagna	9.3	9.7	9.8	–0.4	–0.5
Toscana	6.7	7.7	7.0	–1.0	–0.3
Umbria	4.9	6.3	5.2	–1.4	–0.3
Marche	5.8	6.6	5.0	–0.8	0.9
Lazio	8.3	7.1	6.9	1.2	1.4
Abruzzo	6.0	4.9	3.6	1.2	2.4
Molise	4.0	3.5	1.6	0.5	2.4
Campania	1.7	1.1	0.7	0.6	1.0
Puglia	2.4	2.4	0.2	0.0	2.2
Basilicata	3.3	2.5	1.4	0.8	1.9
Calabria	0.0	0.0	0.3	0.0	–0.3
Sicilia	0.3	0.9	0.0	–0.7	0.3
Sardegna	4.4	3.5	2.7	0.9	1.7

*Regions are ordered from their geographical position, from North to South of Italy

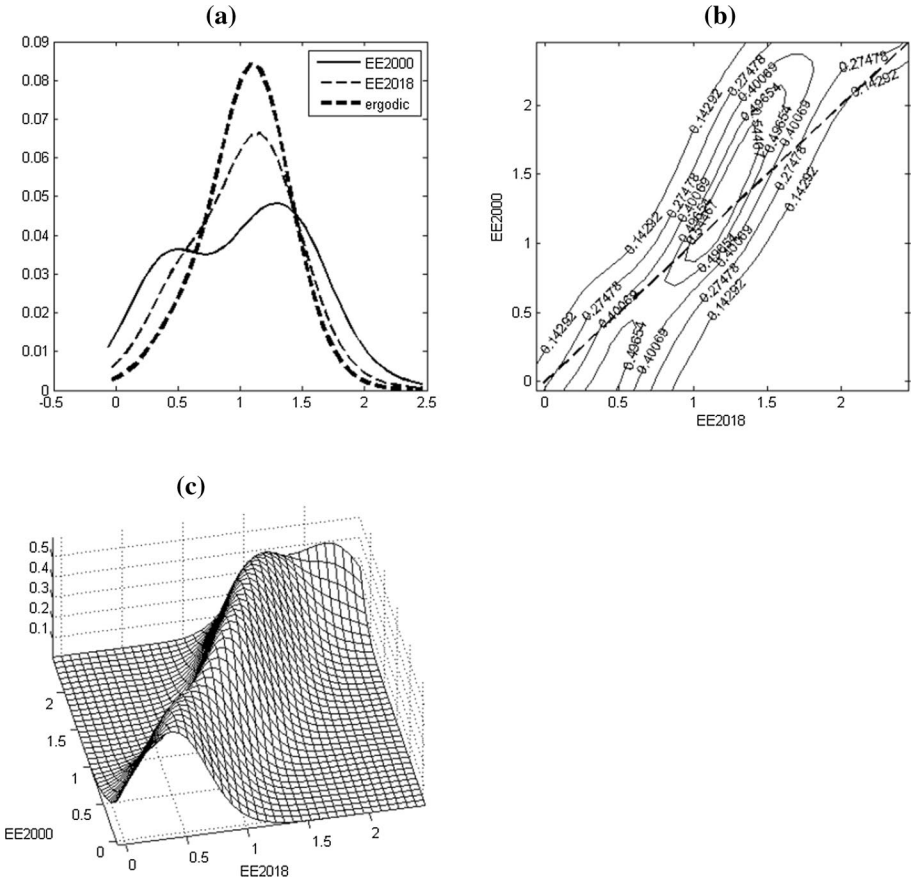


Fig. 7 **a** Initial, final and long-run (ergodic) EE relative distributions, **b** stochastic kernel contour plot **c**, and three-dimensional surface plot of relative EE distributions

geographical proximity cannot explain the disappearance of the polarization pattern in the EE index given that the external shape of the 2018 distribution remains unimodal as in Fig. 7a.

However, geographical proximity exerts an important impact on the transition dynamics of the EE index. Indeed, Fig. 8b, c show a more accentuated counter-clockwise rotation especially above the main diagonal line with the richest provinces worsening. This means that the probability of the richest provinces to move down in the EE index distribution are limited by the existence of spatial effects. In other words, provinces with a high level in the EE index gained from geographical proximity, in the sense that for these provinces geographical proximity dampens the observed decline of the EE index toward lower levels.

There is also a counter-clockwise rotation below the main diagonal at around 0.5 times the average EE index with the poorest provinces improving their position, but this involves lower probability mass. This means that geographical proximity was less beneficial for provinces with a low level of the EE index. One possible explanation is that these provinces are mainly located in the South of Italy and therefore are surrounded by provinces that have similarly low values in the EE index.

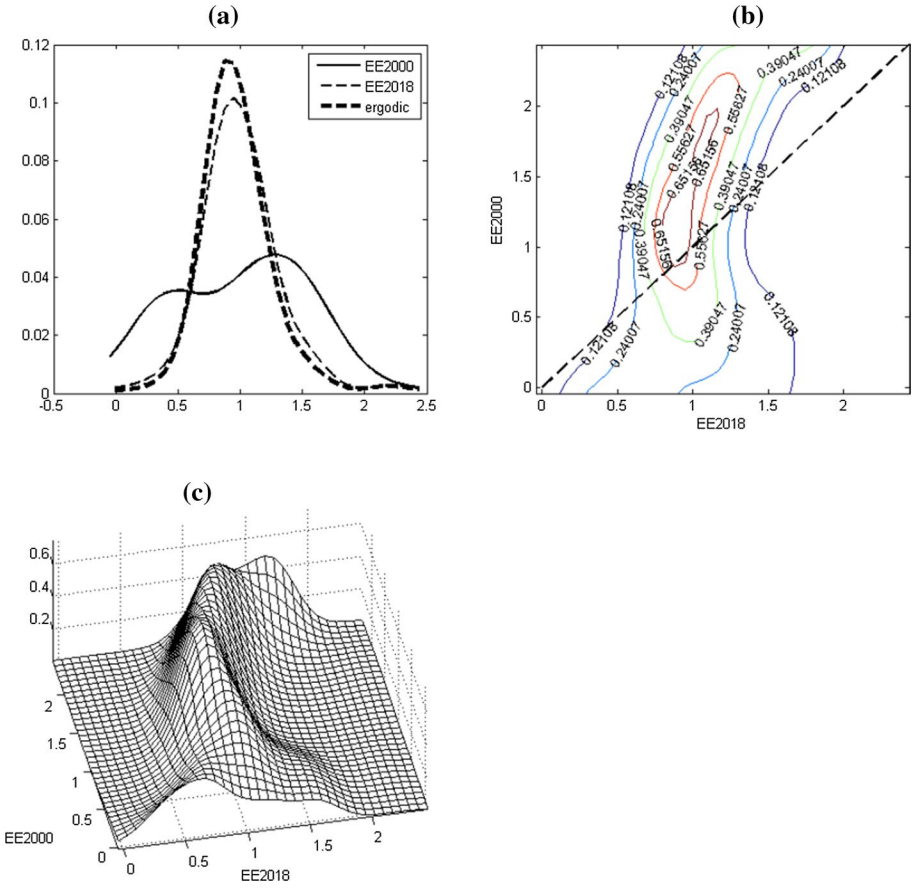


Fig. 8 **a** Space conditioning. Initial, final and long-run (ergodic) EE relative distributions, **b** stochastic kernel contour plot and **c** three-dimensional surface plot of relative EE distributions

4.2 Robustness check: spatial correlation and regression analysis

Previous results show that from around the onset of the financial crisis in 2008 there has been a general deterioration in the mean value of the EE index with some intra-distribution mobility of provinces that produced a convergence process toward the mean. Most importantly, the analysis at the NUTS-3 provincial level highlighted some results that would have been ignored at the NUTS-2 regional level of analysis. Indeed, we found that over time EE variability decreased across provinces but increased within regions.

To further analyze the spatial pattern of the EE index we now evaluate the degree of spatial correlation using data at NUTS-3 level and see how its magnitude changes when using data at NUTS-2 level. We first compute Moran's I statistic (Moran, 1948) and then we estimate the model in Eq. (1). For both analyses, we select the year 2013 since data for

the dependent variable, i.e., the EEBR, the employer enterprise birth rate, in the regression analysis discussed below are not available for more recent years. However, results do not show major changes when a different year is selected.

The Moran I test shows positive spatial autocorrelation of the EE index as most of the observations are in the HH and LL quadrants and are concentrated along the linear fit, for both the provincial (Fig. 9a) and regional (Fig. 9b) level data.[16] This means that provinces/regions with very relatively high/low EE values tend to be located near other provinces with high/low EE values more often than would be expected as a result of purely random factors. However, the spatial correlation is higher at provincial level (0.773) than at regional level (0.552), suggesting that clusters with similar values of EE index are spatially correlated at a more disaggregated geographical level. Similar differences in the spatial correlation emerge when considering a different year.[17]

The result of the Moran scatter plot can also be represented in a map as in Fig. 10a (at provincial level) and in Fig. 10b (at regional level). These maps show provinces/regions with a significant Moran statistic classified by type of spatial correlation: red for the HH association, blue for LL. When the Moran statistic is not statistically significant the provinces/regions are coloured in light grey (Anselin, 1995). Overall, the two maps in Fig. 10 show that the two spatial correlated clusters are located one in the North of Italy, with provinces/regions characterized by a high value of the EE index, and the other in the South of Italy, where provinces/regions with lower level of the EE index are surrounded by provinces/regions with low EE values. The maps also show that the appearance of statistically significant clusters is more dependent on the selection of the spatial unit. For instance, in the South of Italy the Low-Low cluster from the provincial level is geographically more widespread than what is observed at regional level, while in the North of Italy the High-High cluster observed at regional level disappears in some provinces but appears in others. This is evidence of a higher degree of spatial correlation in the EE index when considering NUTS-3 provinces than at NUTS-2 regional level.

We now estimate model (1) to quantify the spatial dependency that has been identified by Moran's I and possible spatial spillovers at both NUTS-3 provincial and NUTS-2 regional level. Table 6 shows estimation results at different levels of spatial aggregation when the dependent variable is the EEBR, the employer enterprise birth rate: column 1–4 provides estimates at NUTS-3 provincial level, while column 5–8 at NUTS-2 regional level.[18]

When we estimate the model with no controls other than time dummies, the choice of the scale aggregation of the data matters. Indeed, the EE index has a larger impact on firm birth at provincial level. However, the MAUP effect diminishes when we include the control variables and when we estimate the spatial models. The elasticities calculated on the EE coefficients are similar to the provincial and regional data, although in the former case these are of larger magnitude. Moving to the spatial model specifications, the neighborhood effect in the SAR model (ρ) is similar across level of

[16] In order to keep the results more comparable across spatial units, we do not use a distance based matrix, but we select at both provincial and regional level a queen first-order contiguity weighting matrix. This spatial matrix selects clusters that shared the same borders. The two islands, i.e., the region of Sardinia and Sicily, are excluded when the analysis is conducted at regional level. Results are robust to different weighting matrices.

[17] Results are available from the authors upon request.

[18] Similar results are found with the number of high growth enterprises as dependent variable. These results are available from the authors upon request.

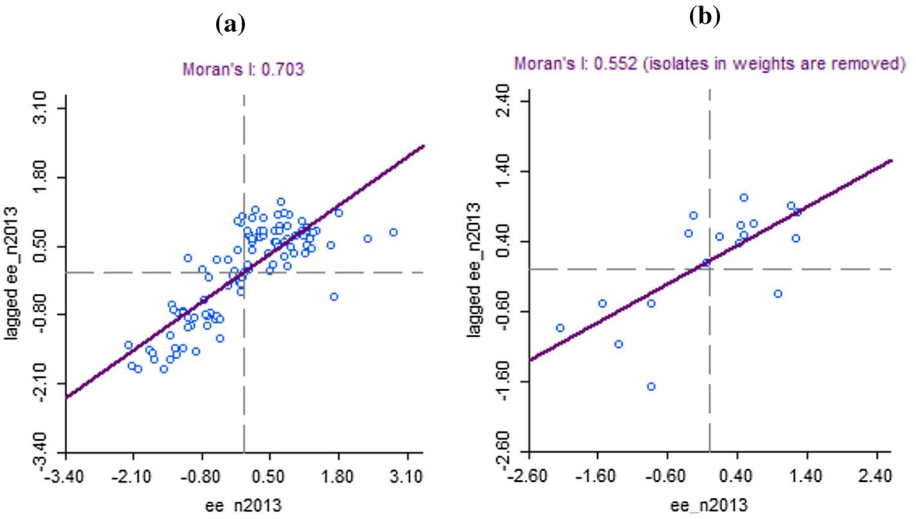


Fig. 9 **a** Moran's I scatterplot on EE index in 2013 at NUTS-3 provincial level **b** and at NUTS-2 regional level

aggregation. However, the indirect (spatial spillover) effects are positive and statistically significant suggesting that a change in the EE index of a neighboring provinces/regions has an impact on the dependent variable. These spatial spillovers are slightly larger in the case of NUTS-3 provinces than in the NUTS-2 case, thus partially confirming the higher spatial correlation detected through Moran's I with smaller geographical units. Interestingly, the values of the coefficients of the spatially lagged error terms (λ) are higher when the SEM is estimated with regional data, suggesting that other explanatory variables, which have been omitted from the model, could be relevant for firm growth when we consider larger spatial units. Still, the EE elasticities have similar magnitude in both NUTS-3 and NUTS-2 cases.

Overall, the robustness exercise provides evidence of strong spatial correlation and statistically significant spatial dependence in the data. It is well-known that the presence of spatial dependence should be taken into account in model specifications because it invalidates linear regression via OLS estimation (Anselin, 2002). Besides this methodological issue, from a theoretical point of view these results emphasize the spatial concept of EE as a set of interconnected actors and may explain why in certain places the conditions that promote entrepreneurial activities are more favorable as they are also affected by spatial spillovers. A number of studies have explored regional variation in entrepreneurship activity based on the hypotheses that the conditions that stimulate such activities differ between locations (see for instance Armington & Acs, 2002; Iacobucci & Micozzi, 2014). However, spatial dimensions have not been incorporated into empirical studies that look at the determinant of EE or how an EE may impact on entrepreneurial activities.

The above results also reveal that these conditions are more spatially correlated at NUTS-3 provincial level but as observed by Arbia (2012), this may be also due to statistical reasons, i.e., the level of spatial correlation generally increase with smaller spatial scale units. However, this argument is controversial in the literature. Studies on agglomeration economies (Burger, M.J. van Oort, F.G. van der Kamp 2010) or on regional economic growth (Dapena et al., 2016) found results to differ across geographical scales mainly

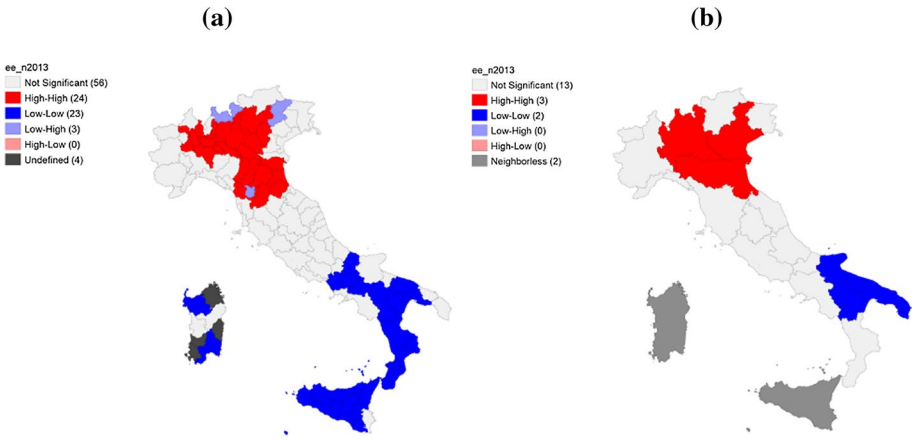


Fig. 10 **a** Moran's cluster map on EE index in 2013 at NUTS-3 provincial level **b** and at NUTS-2 regional level

because the interactions among socio-economics variables and actors vary at different spatial units, so further investigation is also needed in the EE field.

5 Discussion

We believe that the findings have important implications and may contribute to advancing the research on EEs for several reasons. First, the paper provides a measure for assessing how the conditions that shape EEs develop. Indeed, measurement is the most relevant and preliminary question that needs to be answered before conducting any analysis on EEs (Audretsch et al., 2021; Credit et al., 2018; Sternberg et al., 2019). Measurement should involve both EE dimensions, i.e., spatial and dynamic (Bruns et al., 2017; Sternberg et al., 2019) and be made with standard means that allow to overcome EEs heterogeneity and complexity (Shwetzer et al., 2019) while enabling comparative analysis. This can advance our understanding of what constitutes EEs and their impact on entrepreneurial behaviour and economic development and vice versa.

The paper also takes the analyses at a low level of spatial disaggregation, i.e., the NUTS-3 provinces. This is a novelty in the EE literature. Some of the previous studies identified and measured EEs mainly at national level (see, for instance, Ács et al., 2014; Isenberg, 2011; Stangler J. Bell-Masterson 2015), although in recent years there have been some attempts in the direction of implementing the national measures to smaller scale. For instance, the GEM NECI (National Entrepreneurship Context Index) from the GEM (Global Entrepreneurship Monitor) or the REDI (Regional Entrepreneurial Development Index) from the national GEI (Global Entrepreneurship Index), but these indicators refer to regions (NUTS-1 or NUTS-2) or countries.[19] Other measures of EEs at NUTS-2 level

[19] For more details see http://www.gemconsortium.org/images/media/gem-neci-background-notes-1613753990.pdf for the GEM NECI and https://thegedi.org/regional-gedi/ and Ortega Argiles et al., (2014) for the REDI.

Table 6 Random effects and spatial panel estimation (2000–2013). Dependent variable: Employer enterprise birth rate (EEBR)

	NUTS-3 Provincial level				NUTS-2 Regional level			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables	RE	RE	SAR	SEM	RE	RE	SAR	SEM
EE Index	0.427***	0.374***	0.334***	0.346***	0.331***	0.344***	0.274***	0.278***
	(0.039)	(0.043)	(0.041)	(0.043)	(0.078)	(0.091)	(0.087)	(0.077)
Trade openess over VA		-0.255	-0.281	-0.257		-1.357	-2.095*	-1.007
		(0.296)	(0.290)	(0.282)		(1.226)	(1.176)	(1.363)
Industry heterogeneity index		9.471***	8.359***	8.401***		-0.943	-0.051	-0.334
		(2.470)	(2.451)	(2.442)		(1.179)	(1.124)	(1.214)
Population (log)		-0.375**	-0.308	-0.324		-0.164	-0.002	-0.041
		(0.186)	(0.213)	(0.210)		(0.117)	(0.146)	(0.100)
Constant	8.378***	5.263*	2.963	5.638*	8.784***	12.012***	6.588***	9.918***
	(0.197)	(2.792)	(3.094)	(3.092)	(0.437)	(1.822)	(2.234)	(1.888)
ρ			0.252***				0.273***	
			(0.042)				(0.064)	
λ				0.260***				0.462***
				(0.043)				(0.174)
EE elasticity	0.243***	0.212***		0.197***	0.213***	0.222***		0.180***
	(0.021)	(0.024)		(0.043)	(0.050)	(0.059)		(0.080)
Spatial effects of EE								
Direct effects			0.341***				0.284***	
			(0.043)				(0.093)	
Indirect effects			0.109***				0.091**	
			(0.028)				(0.042)	
Observations	1,442	1,442	1,442	1,442	280	280	280	280
R-squared	0.616	0.608	0.583	0.612	0.754	0.743	0.968	0.758

Table 6 (continued)

	NUTS-3 Provincial level				NUTS-2 Regional level			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of units	103	103	103	103	20	20	20	20
Time dummies	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Cluster standard errors at NUTS-3 and NUTS-2 level in parentheses $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$

are limited (Guzman & Stern, 2020; Stam, 2015; Sternberg et al., 2019) and are absent at more disaggregated territorial level such as the NUTS-3 provincial level. The only exemption is Stam (2018), who builds an EE index for the Netherland provinces, which however is equivalent to the NUTS-2 according to the EU classification, and Guzman and Stern (2020) who provide a framework for assessing the EE at the US national and state level. More recently, Sternberg et al., (2019) developed an EE index based on survey data at NUTS-2 regional level for Spain and Germany following the EE concept of Stam (2018). There are also single case studies and some analysis of emerging ecosystems at city level (Cohen, 2006; Mack & Mayer, 2016; Motoyama & Knowlton, 2017; Pugh et al., 2019; Spigel, 2017), but their diversity creates serious limitations in terms of EE comparison.

Researchers considered the lack of studies at a smaller scale unit than the national level as a weakness in the EE literature, given that most of the factors that characterize EEs, vary largely between regions, provinces and, in some cases, cities. As noted by Sternberg et al. (2009, p.215) "entrepreneurs need the primarily regional, informal network of friends, former colleagues and bosses, relatives and first customers to be able to realize their start-up idea at all, particularly at the first attempt. It is regional or even local institutions such as banks, chambers industry and commerce and heads of business incubators who give "entrepreneurs-to-be" the decisive stimuli in this early phase of the entrepreneurial process". Indeed, the existing literature has recognized EE as a spatially bounded entity that follows a strong spatial logic. It is argued that close geographic proximity fosters network formation and knowledge exchange (Glückler, 2007) and that entrepreneurial activity cannot be seen as independent from the environment within which businesses operate; rather the community and the culture of a given place can have a significant impact on entrepreneurship (Autio et al., 2014). For instance, universities and research centres with their traditional mission of knowledge transfer and knowledge spill-over play a relevant role for the development of EE.

The university's entrepreneurial role based on third mission is also important to disseminate knowledge especially in the surrounding area where universities and research centres are located (Baldini et al., 2015; Bolzani et al., 2014; Chiarello et al., 2021). The articles in a recent special issue provide important insights regarding the ways in which knowledge transfer takes place in Italy (Grimaldi et al., 2021). For instance, Villani and Lechner (2021) argue that universities can create a community of specialized and complementary actors and form a supportive network in the local ecosystem, while Barbini et al. (2021) explore the ways universities promote student entrepreneurship. The university's entrepreneurial role is also analysed by Sciarelli et al. (2021), as they focused on university spin-offs in the South of Italy. Therefore, the role of local institutions, organizations, universities, and the connections between the elements that compose an EE matter (Motoyama & Knowlton, 2016). Indeed, the term ecosystem has become prevalent among researchers to describe the interactions between economic agents and their environment (Acs et al., 2017). The literature also acknowledges that regional agglomerations of local factors and resources and certain mechanisms that shape the EE exist outside its boundaries, and that spatial spillovers in terms of knowledge, institutional policy, social interactions, and networks of entrepreneurs are facilitated through geographic proximities (D. B. Audretsch et al., 2019). There is also empirical evidence that the level and type of entrepreneurial activities varies systematically across different EEs, which in turn differ in several elements among clusters of spatial units (Content et al., 2020). As a result, the analysis at a lower level of spatial disaggregation is extremely important.

Third, the paper contributes to the literature by examining the dynamic components of the EE. Different indicators, proxies, and measures, such as the GEM or the REDI have

been used to explore the time evolution of EEs. The problems with most of these country level or NUTS-1 level indices is that they rely heavily on the availability of data and ignore the specificities needed to analyse EEs on a smaller spatial scale (Brown & Mason, 2017). Studies at city level have tried to overcome these limitations, by offering a detailed description of how entrepreneurial activities take place, however they mainly take a static view of the concept, and their diversity creates serious limitations in terms of EE comparison (Sternberg et al., 2019). Spigel (2017) observed that, in an EE, elements can develop simultaneously and reinforce each other, though they can never replace one another completely. Others claim that the elements of EEs will shift in importance and in their connections as they evolve (Mack & Mayer, 2016). In this paper we try to reconcile the need to measure EEs at a small spatial scale unit with the necessity to understand how EEs evolve over time.

Fourth, the analysis is carried over by focusing on the Italian case, thus it provides a contribution to the arguments advanced in the literature for explaining the heterogeneous entrepreneurial performance (Iacobucci & Micozzi, 2014) and differences in economic growth observed at the Italian provincial level. Indeed, the analysis for the Italian case showed that on average the conditions that make up the EE and that foster productive entrepreneurship are unevenly geographically distributed and have deteriorated after 2008, especially among the richer (with higher values of the EE index) provinces, which are mainly located in the North of Italy.

The paper also contributes to the design and implementation of the policy-making process. Indeed, local authorities may contribute to shape and promote the EE, given that their policies may directly or indirectly impact on the EE components. Moreover, the EE index may complement other statistical tools such as the Regional Innovation Scoreboard (RIS) and help policy makers to target policies designed at fostering innovation at regional or provincial level. In the Italian case, provincial authorities have a limited importance in the administrative structure, however they are responsible for implementing decentralized policies in relevant sectors such as health, education, transport, culture, energy policy and environmental protection, which are functional in promoting and sustaining the economic environment and in creating the necessary conditions for firms, universities, and entrepreneurs to implement entrepreneurial activities (Calcagnini & Perugini, 2019a; Veneri & Murtin, 2016). Indeed, policy measures on land-use, transportation, human capital, and housing, may be designed and implemented in different ways within the same region or province. Therefore, the adoption of targeted measures aimed at expanding the quality and the quantity of infrastructures and business services, at increasing the supply and accessibility of finance and investment in R&D, and at fostering the endowments of social and human capital are essential for enhancing firm competitiveness and for reducing the divergence that exists between the North and the South of Italy in terms of EE conditions, entrepreneurship growth rates, and economic development (Calcagnini & Perugini, 2019a; Iacobucci & Micozzi, 2014; Putnam, 1993; Toniolo 2013).

6 Limitations

The paper has some limitations, which give rise to several opportunities for future research. First, the quantification of the EE has been constrained by data availability. More direct measures of EEs in terms of entrepreneurial attitudes, abilities, and aspirations have not been taken into account because they are hardly measured at provincial level. Therefore,

it would be interesting to complement the measuring of EEs with survey data to capture these features.

A second limitation is in the use of variables able to capture technological innovation, which has an increasing and profound impact on entrepreneurship and venture creation. Only recently has the EE paid more attention to describing and understanding how digital technologies may influence the nature and interactions among actors and the components of the EEs.

A third limitation of the paper concerns its nature, which is mainly descriptive. However, this may represent an avenue for future research, for instance to examine how EEs contribute to explaining performance differences across territories in terms of creation of firms and economic growth. A related interesting issue would be to assess how, and through which channels, EEs may impact on other entrepreneurial variables, linked for instance to the creation of high-growth firms or the industrial structure of the territories. Indeed, the causal relations between EEs and entrepreneurship have not yet been studied sufficiently (Alvedalen & Boschma, 2017; Stam & Spigel, 2018). Another interesting issue would be to analyse how the various components of an EE interact with each other and what is their impact on entrepreneurial activities. The use of longitudinal data would help researchers to investigate these issues with a more efficient empirical instrument.

7 Conclusions

Interest in the concept of an entrepreneurial ecosystem (EE) has been growing in recent years. An EE can be defined in several different ways, but most of them highlight the combination or interaction of elements producing shared innovative and cultural values that support entrepreneurial activities within a territory (Cao & Shi, 2020; Cavallo et al., 2019; De Brito & Leitão, 2021; Malecki, 2018; Wurth et al., 2021). Even though the concept has quickly gained popularity and despite the growing number of published works, there is a lack of studies that measure EE at regional or provincial level and combine the spatial and the dynamic view of EE (Alvedalen & Boschma, 2017; Stam and van de Ven 2021; Sternberg et al., 2019). To date, scholars have mainly considered EEs as a static concept and have primarily proposed the analysis at national level (see, for instance, Ács et al., 2014; Isenberg, 2011; Stangler J. Bell-Masterson 2015). Some qualitative case studies combined both the spatial and the time dimensions (Cohen, 2006; Mack & Mayer, 2016; Pugh et al., 2019; Spigel, 2017), however, although they are useful to understand the interconnections and the role played by certain actors in a single EE, their diversity creates serious limitations in terms of EE comparison.

The paper tries to fill these gaps, i.e., (1) it proposes a rigorous spatial unit to analyze EEs, (2) it measures EEs at this spatial scale level and in a dynamic perspective, (3) it analyzes how EEs and the conditions that make up the EEs evolve over time and across spatial units, and then (4) it assesses the sensitivity of the EE index to changes in the scale of the spatial unit, i.e., whether it is more appropriate to measure and examine the EEs at NUTS-3 provincial level or NUTS-2 regional level.

The analysis considers the 2000 to 2018 period, and it is carried out at the Italian NUTS-3 provincial level, which we believe is the most appropriate spatial unit to identify and measure EEs. Indeed, our findings support the theoretical arguments advanced in the EE literature about the need to analyze such ecosystems at a more disaggregated territorial level than the regional or the national level. To measure EEs, we follow the framework proposed by Stam (2015) and construct an index of EEs, which is a novelty at the Italian NUTS-3 provincial level, by modifying the version of Iacobucci and Perugini (2020a, 2020b), adjusted to account for the temporal dimension of the data.

To examine the evolution of EEs across time and space the paper proposed statistical tools and regression analysis. These were used to assess inequality in the EE index through the Theil's entropy measure (Theil, 1971), which also allowed to determine the contribution to inequality by considering the level of EEs in the neighbouring provinces, that is provinces that belongs to the same NUTS-2 region. We also explored the time dimension of the data and assessed how the degree of heterogeneity in the EE index has changed over time but, most importantly, what is the dispersion of the provincial EE within the larger NUTS-2 region. Statistical evidence was also provided in terms of spatial correlation and spatial dependence in the data to assess whether the spillover effects are more relevant at provincial or regional level, thus providing an indication of which measure of spatial unit is more adequate to account for the strength of the interconnections that exist between the EE components. Only recently has the entrepreneurship literature recognized the relevance of incorporating spatial modelling techniques for such analysis (Plummer, 2010), however to our knowledge these have not been applied in the EE context yet.

Spatial analysis and spatial models are necessary because the existence of spatial dependence renders ordinary least square estimation inappropriate. In addition to the spatial analysis, the paper proposed an analysis on the time evolution of the cross-sectional distribution of the EE index, which is investigated through the distribution dynamic approach developed by Quah (1993, 1996, 1997). This approach provides useful insights to whether the value of the EE in a province has changed, if there has been a convergence process of the EE index across provinces, i.e., if the EE levels in poorer (with a low endowment of the EE index) provinces are catching up in relative terms with those in the rich ones (with a high EE index endowment), or if the convergence occurs within two or more clusters of homogeneous provinces, i.e., if the distribution of the EE index is polarized. Like the spatial analysis approach, which is extensively used in the regional science literature, the distribution dynamic approach has not been employed in the EE literature yet but offers a different perspective to understand how the provincial EEs evolved over time but also across the entire EE distribution, and to evaluate how specific factors, such as for instance geographical proximity, may have an impact on the transition dynamic in the value of the provincial EE.

Results show the existence of a heterogeneous geographical distribution of the EE index with two distinct spatially correlated clusters, one located in the more developed Northern part of Italy, with provinces characterized by high values of the EE index, and the other in the South of Italy, where provinces with lower levels of the EE index are surrounded by provinces/regions with low EE values. This feature does not disappear over time, although there is a tendency for an overall deterioration of the mean value of the EE index toward lower levels. Findings also support the choice of selecting NUTS-3 province as a spatial unit of analysis and confirm the theoretical arguments typically advanced in the literature that view EE as a complex system that emerges and evolves at local level. Indeed, results reveal an increasing within regional variability in the EE index, spatial correlation in the data, and some spatial patterns in the EE index that would have been ignored using larger spatial units.

The dynamic perspective adopted in the paper also shows some intra-distribution mobility of provinces (the poor provinces becoming richer and the rich provinces becoming poorer) that produced a convergence process toward the mean value of the EE index and the disappearance of the polarization pattern that emerged at the beginning of the period. Results also show that geographical proximity is a factor that has limited the extent of this convergence process.

In this paper we addressed two main shortcomings in the EEs literature that are at the center of the recent debate and that deals with the measurement and the evolution of EEs, i.e., the spatial dimension and the time dimension of EEs, so we hope this paper could encourage scholars to further investigate these weakness in a future research agenda on EE.

Appendix

(See Table 7 8 and 9).

Table 7 The EE index in 2000 and 2018

NUTS-3 code	NUTS-3 provinces	NUTS-2 code	NUTS-2 regions	NUTS-1	EE (2000)	EE (2018)
15	Milano	3	Lombardia	North-West	10.0	10.0
37	Bologna	8	Emilia-Romagna	North-East	8.0	8.4
58	Roma	12	Lazio	Centre	6.3	7.9
34	Parma	8	Emilia-Romagna	North-East	8.4	7.8
1	Torino	1	Piemonte	North-West	6.9	7.2
28	Padova	5	Veneto	North-East	7.1	7.2
36	Modena	8	Emilia-Romagna	North-East	7.1	7.1
35	Reggio nell'Emilia	8	Emilia-Romagna	North-East	8.5	7.1
21	Bolzano/Bozen	4	Trentino-Alto Adige/Südtirol	North-East	6.4	6.9
3	Novara	1	Piemonte	North-West	6.9	6.8
40	Forlì-Cesena	8	Emilia-Romagna	North-East	6.7	6.8
4	Cuneo	1	Piemonte	North-West	5.3	6.5
99	Rimini	8	Emilia-Romagna	North-East	7.3	6.5
17	Brescia	3	Lombardia	North-West	7.4	6.5
23	Verona	5	Veneto	North-East	6.5	6.4
33	Piacenza	8	Emilia-Romagna	North-East	5.7	6.3
93	Pordenone	6	Friuli-Venezia Giulia	North-East	5.9	6.3
22	Trento	4	Trentino-Alto Adige/Südtirol	North-East	6.1	6.2
48	Firenze	9	Toscana	Centre	6.2	6.1
100	Prato	9	Toscana	Centre	6.4	6.1
16	Bergamo	3	Lombardia	North-West	7.2	6.1
6	Alessandria	1	Piemonte	North-West	5.8	6.0
5	Asti	1	Piemonte	North-West	5.1	6.0
13	Como	3	Lombardia	North-West	6.4	5.8
24	Vicenza	5	Veneto	North-East	6.9	5.8
12	Varese	3	Lombardia	North-West	6.0	5.7
50	Pisa	9	Toscana	Centre	5.8	5.6

Table 7 (continued)

NUTS-3 code	NUTS-3 provinces	NUTS-2 code	NUTS-2 regions	NUTS-1	EE (2000)	EE (2018)
39	Ravenna	8	Emilia-Romagna	North-East	6.3	5.6
96	Biella	1	Piemonte	North-West	4.6	5.6
38	Ferrara	8	Emilia-Romagna	North-East	5.1	5.6
97	Lecco	3	Lombardia	North-West	6.1	5.5
26	Treviso	5	Veneto	North-East	6.4	5.5
20	Mantova	3	Lombardia	North-West	6.0	5.5
98	Lodi	3	Lombardia	North-West	6.2	5.5
19	Cremona	3	Lombardia	North-West	5.6	5.4
30	Udine	6	Friuli-Venezia Giulia	North-East	5.4	5.4
7	Valle d'Aosta/Vallée d'Aoste	2	Valle d'Aosta/Vallée d'Aoste	North-West	6.2	5.2
46	Lucca	9	Toscana	Centre	4.4	5.1
10	Genova	7	Liguria	North-West	4.7	5.1
18	Pavia	3	Lombardia	North-West	5.0	5.1
42	Ancona	11	Marche	Centre	4.5	5.1
51	Arezzo	9	Toscana	Centre	5.0	5.1
32	Trieste	6	Friuli-Venezia Giulia	North-East	5.2	5.1
11	La Spezia	7	Liguria	North-West	4.4	5.0
29	Rovigo	5	Veneto	North-East	4.1	4.9
45	Massa-Carrara	9	Toscana	Centre	4.7	4.9
41	Pesaro e Urbino	11	Marche	Centre	4.8	4.9
2	Vercelli	1	Piemonte	North-West	5.0	4.9
9	Savona	7	Liguria	North-West	4.6	4.9
27	Venezia	5	Veneto	North-East	5.6	4.8
44	Ascoli Piceno	11	Marche	Centre	4.1	4.8
47	Pistoia	9	Toscana	Centre	4.9	4.7

Table 7 (continued)

NUTS-3 code	NUTS-3 provinces	NUTS-2 code	NUTS-2 regions	NUTS-1	EE (2000)	EE (2018)
52	Siena	9	Toscana	Centre	4.6	4.5
31	Gorizia	6	Friuli-Venezia Giulia	North-East	5.6	4.5
103	Verbano-Cusio-Ossola	1	Piemonte	North-West	5.7	4.5
66	L'Aquila	13	Abruzzo	South and Isles	2.0	4.4
49	Livorno	9	Toscana	Centre	4.6	4.4
68	Pescara	13	Abruzzo	South and Isles	3.7	4.4
14	Sondrio	3	Lombardia	North-West	4.2	4.3
8	Imperia	7	Liguria	North-West	3.4	4.3
54	Perugia	10	Umbria	Centre	4.1	4.2
55	Terni	10	Umbria	Centre	3.3	4.1
62	Benevento	15	Campania	South and Isles	1.8	4.1
69	Chieti	13	Abruzzo	South and Isles	2.7	4.0
43	Macerata	11	Marche	Centre	3.2	4.0
67	Teramo	13	Abruzzo	South and Isles	3.6	4.0
72	Bari	16	Puglia	South and Isles	2.0	4.0
53	Grosseto	9	Toscana	Centre	3.8	4.0
56	Viterbo	12	Lazio	Centre	3.2	3.9
92	Cagliari	20	Sardegna	South and Isles	2.7	3.8
25	Belluno	5	Veneto	North-East	3.7	3.7
59	Latina	12	Lazio	Centre	4.4	3.5
65	Salerno	15	Campania	South and Isles	2.3	3.4
94	Isernia	14	Molise	South and Isles	2.5	3.3
64	Avellino	15	Campania	South and Isles	2.3	3.3
91	Nuoro	20	Sardegna	South and Isles	1.9	3.2
70	Campobasso	14	Molise	South and Isles	1.5	3.2
90	Sassari	20	Sardegna	South and Isles	3.3	3.1

Table 7 (continued)

NUTS-3 code	NUTS-3 provinces	NUTS-2 code	NUTS-2 regions	NUTS-1	EE (2000)	EE (2018)
76	Potenza	17	Basilicata	South and Isles	2.3	3.1
60	Frosinone	12	Lazio	Centre	2.0	3.0
77	Matera	17	Basilicata	South and Isles	1.5	3.0
61	Caserta	15	Campania	South and Isles	1.7	2.5
75	Lecce	16	Puglia	South and Isles	1.2	2.5
63	Napoli	15	Campania	South and Isles	1.6	2.4
87	Catania	19	Sicilia	South and Isles	1.5	2.3
71	Foggia	16	Puglia	South and Isles	0.5	2.3
74	Brindisi	16	Puglia	South and Isles	0.9	2.2
95	Oristano	20	Sardegna	South and Isles	1.4	2.1
88	Ragusa	19	Sicilia	South and Isles	2.0	2.1
79	Catanzaro	18	Calabria	South and Isles	2.3	2.1
57	Rieti	12	Lazio	Centre	2.6	2.1
89	Siracusa	19	Sicilia	South and Isles	1.9	2.0
78	Cosenza	18	Calabria	South and Isles	1.6	2.0
83	Messina	19	Sicilia	South and Isles	0.7	1.9
73	Taranto	16	Puglia	South and Isles	1.1	1.8
82	Palermo	19	Sicilia	South and Isles	1.2	1.7
81	Trapani	19	Sicilia	South and Isles	1.0	1.6
102	Vibo Valentia	18	Calabria	South and Isles	1.1	1.0
80	Reggio di Calabria	18	Calabria	South and Isles	0.2	0.7
85	Caltanissetta	19	Sicilia	South and Isles	0.6	0.7
86	Enna	19	Sicilia	South and Isles	0.2	0.5
84	Agrigento	19	Sicilia	South and Isles	0.0	0.4
101	Crotone	18	Calabria	South and Isles	1.3	0.0

NUTS-3 provinces are ranked according to EE (2018)

Table 8 Variables used for model estimation

Variables	Definition	Source	Year
EEBR	Birth of an enterprise that has at least one employee in the birth year and of an enterprise that existed before the year in consideration over adult population	ASIA-ISTAT	2000–2013
EE index	Entrepreneurial Ecosystem index	Various sources	2000–2013
Trade openness	(Import + Export)/Value Added	ISTAT	2000–2013
Industry heterogeneity	Inverse of the Herfindahl–Hirschman index calculated on the number of active firms	Movimprese	2000–2013
Population (log)	Number of resident populations	ISTAT	2000–2013

Table 9 Summary statistics of model variables

Variables	Obs	Mean	Min	Max	p50	sd
NUTS-2 Province						
EEBR	1442	8.291	18.138	4.114	8.066	2.134
EE index	1442	4.720	10.000	0.000	5.129	2.384
Trade openness	1442	0.030	0.470	-0.785	0.026	0.157
Industry heterogeneity	1442	0.882	0.942	0.704	0.893	0.042
Population	1442	12.933	15.246	11.374	12.847	0.708
NUTS-3 Regions						
EEBR	280	8.389	12.782	4.340	8.300	1.907
EE index	280	5.424	10.000	0.000	6.111	3.153
Trade openness	280	0.026	0.228	-0.182	0.032	0.084
Industry heterogeneity	280	0.948	0.991	0.776	0.964	0.046
Population	280	14.462	16.106	11.688	14.403	1.061

(1) Building the Entrepreneurial Ecosystem (EE) index

Our measure of the EE follows the theoretical framework proposed by Stam (2018) and builds an index of EEs at the NUTS-3 provincial level by modifying the version of Iacobucci and Perugini (2020a, 2020b), adjusted to account for the temporal dimension of the data. Stam's (2018) framework is focused on the conditions of a region to promote and sustain EEs. Accordingly, the EE is viewed as the interaction of systemic conditions and framework conditions. The former, such as networks of entrepreneurs, leadership, finance, talent, knowledge, and support services, are at the heart of Stam's (2018) EE, while the latter entail a social context (informal and formal institutions) that enables or constrains human interaction.

Iacobucci and Perugini (2020a, 2020b) applied this framework to the Italian case and built the EE for the year 2011. In this paper we extend this analysis to the 2000–2018 period and build the EE following the conditions selected by Stam (2018). However, in the construction of the index we also take into account how the domains and features characterizing EEs have been selected by the literature (Isenberg, 2011; Mason C.& Brown 2014; Ortega Argiles et al., 2014; Stangler J. Bell-Masterson 2015; Sternberg et al., 2019). For instance, we include the "Firm survival" condition in our EE index. The conditions are: (1) Social capital; (2) Entrepreneurship Culture; (3) Physical Infrastructure; (4) Value Added; (5) Finance; (6) New knowledge; (7) Business services; (8) Firm survival; (9) Human capital; (10) Migration flow; (11) Population structure. Due to data availability, we dropped formal and informal institutions from Stam's (2018) selection. To proxy these variables/conditions, we gathered data from different sources (see Table 1 in the paper).

(a) Description of the selected variables

(1) Social capital

The literature on social capital and its positive impact on economic development, innovation, productivity, well-being and other socio-economic aspects of life is extensive (see, for

instance, Bjørnskov, 2006, among others). More recently, studies have provided evidence of a beneficial effect on entrepreneurial activities. Indeed, social capital, which includes personal relationships, trust, and civic norms, allows entrepreneurs to benefit from positive externalities (Dana et al., 2019; Pittz et al., 2019; Theodoraki et al., 2018). It is argued that formal or informal relationships with family and friends, or networking between groups of people is particularly important in providing intangible resources and competencies of various kinds, such as technological, commercial, administrative and organisational capabilities, to nascent entrepreneurs (Romano et al., 2017). Also, relations with the community and the territory are equally important because they can facilitate the procurement of financial resources, through mechanisms such as, for instance, crowdfunding (Giudici et al., 2018). The literature also shows that a stable environment, in which trust is an essential element, is favourable to the transfer of knowledge and capabilities among members (Iansiti & Levien, 2004; Theodoraki et al., 2018). When relationships are based on trust, members are more willing to cooperate and exchange information, thus facilitating entrepreneurial activities. Moreover, to build trust within an ecosystem, shared norms of behaviour are also a powerful tool (Thomas & Autio, 2014).

Social capital is defined and measured in various ways (Bourdieu, 1986; Coleman, 1988; Putnam, 1993). For these reasons a precise and widely accepted definition is still not available (Adler & Kwon, 2002). The lack of a common definition of social capital has given rise to different measures and a variety of applied empirical works. Generally, these studies classify social capital into several dimensions, which are then reduced to a single index. In this paper, we construct a measure of social capital taking into account those dimensions that have been identified in the literature to be relevant elements for promoting entrepreneurial activities. More specifically, we normalize using the min–max method and aggregate with the arithmetic mean three variables that are proxy for social networks, civic norms, and trust respectively: a) the number of employees in social cooperatives over total employees; b) the percentage of municipal waste object of separate collection on total municipal waste; and c) the number of protests on ordinary bills, drafts and bank checks over the resident population.[20] The source of this data is ISTAT.

Even though our selection was constrained by the lack of data at NUTS-3 provincial level for the period under observation (2000–2018), the geographical distribution of the obtained social capital index shows the expected results, i.e., provinces located in the North and the Centre of Italy have a greater endowment of social capital than Southern provinces. This result is in line with previous findings (Cartocci, 2007; Micucci & Nuzzo, 2003). For instance, the Spearman rank correlation coefficients of our social capital index and the social capital index computed by Cartocci (2007), a widely cited work in the social capital literature, is 0.69. For robustness exercise we also compute the social capital index with the principal component analysis. Results do not show sensible changes.

2) Entrepreneurship Culture

This variable measures the degree to which entrepreneurship is valued in society (Stam, 2018). The literature often used the share of small businesses or average firm size as a proxy for the entrepreneurial climate (Armington & Acs, 2002; Lee et al., 2004). We use

[20] For more details on the methodology used to construct the social capital index see Section b) Index construction and robustness, in this Appendix.

the number of start-ups per thousands of inhabitants. Data are from *Movimprese*, which is a regular survey of the demography of Italian companies conducted by the *Unioncamere*, the Italian Chamber of Commerce.

3) Physical infrastructure

The important role of transport infrastructure for economic development has been widely investigated in the literature (see, for instance, Audretsch, Heger, et al., 2015; Audretsch, Link, et al., 2015; Spiekermann & Wegener, 2006). It is generally argued that areas with better access to the locations of input materials and markets will be more productive, more competitive and hence more successful than more remote and isolated areas. Our measure of physical infrastructure is an accessibility index. The index takes into account transport infrastructure such as total length of roads, motorways and rail lines in the province itself, but also the connectivity of transport networks, i.e., activities and opportunities that can be reached by the network itself. In other words, we also consider how easily destinations can be reached in terms of travel time, cost or inconvenience. A recent measure of accessibility index for the Italian provinces, which is used in this paper, is provided by Messina (2007).

4) Value Added

The spatial dimension of market location is an influential factor that has an impact on entrepreneurial activities. The relevance of the market may also vary in relation to specific products (Sternberg et al., 2019). Stam (2018) measures demand as a composite index consisting of per capita disposable income, population, and an indicator of potential market demand. In this paper we use per-capita Value Added. Value Added at provincial level is from ISTAT and it is measured at current prices. To deflate Value Added we use the consumer price index, which is measured in the main cities of provinces and in regions.

5) Finance

The supply and accessibility of finance for firms is vital for firms' growth and survival (Ortega Argiles et al., 2014; Parker, 2004). Without a large and efficient credit market to supply firms, some entrepreneurs will face a financial barrier making it impossible to seize opportunities. We use the amount of loans to households and firms as a share of Value Added. Data are from the Bank of Italy and ISTAT.

6) New knowledge

Investments in new knowledge are an important source of entrepreneurial opportunities (Mason C.& Brown 2014). New knowledge can be measured in different ways, from patents, innovation projects or expenditure in R&D. The indicator of new knowledge used in this paper is the capital stock of R&D by public and private firms over Value Added. The National Institute of Statistics (ISTAT) provides only data on R&D expenditure at regional level. To construct the stock of R&D expenditure at provincial level we first update the time series of regional capital stock of R&D from Bronzini and Piselli (2009), using the perpetual inventory method. At regional level, this time series is highly correlated with other economic variables such as the Value Added, the number of graduated students, and

the number of firms in high tech industries. In the worst case, that is in the case of the number of graduated students, the correlation coefficient with the stock of R&D is above 0.65. Therefore, we use the province average shares of these three variables to apportion the regional data of the R&D capital stock. Data on expenditure in R&D at regional level are from ISTAT.

7) Business services

Business services are an important driver of the knowledge-based economy and a key element for establishing enterprises (Feld, 2012). The growing trend in outsourcing has increased the demand for business services, especially for some core activities such as the recruitment services, ICT development, and administrative activities. The availability of such specific intermediate business services can lower the barrier to firm creation (Stam, 2018). We measure intermediate services as the percentage of active firms in the professional, scientific and technical activities (that are within the M section under the Nace Rev. 2 classification of economic activities) and in the administrative and support service activities (N) in the population of active firms in the business service sector (G-O). Data are from *Movimprese*.

8) Firm survival

The capacity of a firm to survive within a defined area is another important element that shapes an EE. Indeed, the empirical literature shows that if firms belong to a vibrant ecosystem than it increases its odds of survival (Ahmad & Hoffman, 2008). We use the survival rate, which is defined as the percentage of start-up firms at time t-3 that survived at time t and the number of start-up firms at time t-3. Data are from ISTAT.

9) Human capital

A relevant explanatory factor of the entrepreneurial activity is to possess the skills and knowledge required to create new companies. The literature shows that these human capital outcomes are vital to discover and create entrepreneurial opportunities, and to assist in the accumulation of new knowledge and the creation of advantages for new firms (Marvel et al., 2016). In the literature it is common to measure human capital with the number of graduated students or with the educational attainment of the resident population. However, work experience it is also assumed to be an important element for entrepreneurial activities. Therefore, in this paper we combine these two elements, and we construct a measure of human capital by considering both formal education and work experience of the resident population. For this purpose, we used the earnings functions of education proposed by Mincer (1974) and estimate the returns to investment in education as follows:

$$\log W_i = \gamma S_i + \psi_1 T_1 + \psi_2 T_i^2$$

Accordingly, the log of the worker's wage ($\log W_i$) in province i depends on the years of schooling (S) and labour experience (T), which enters the equation with an additional quadratic term. The parameters γ measure private returns from investment in education while ψ_1, ψ_2 the return from the accumulated labor experience.[21] The calculation of years of schooling (S_i) is obtained by the number of years required to reach a certain level of qualification, weighted by the relative province educational qualifications.[22] With regard to labor experience (T_i), we follow the specification provided by Bils and Klenow (2000):

$T_i = (Age_i - S_i - 6)(1 - u_i)$.

It is assumed that the years of labor experience are given by the difference between an individual's age (Age) and his/her years of schooling (after subtracting 6 years, since schooling in Italy usually starts at the age of 6). In addition, we add the unemployment rate (u_i) to adjust for the possibilities of not accumulating years of labor experience when one is unemployed. Data for both the years of schooling (S) and the labor experience (T) are available from ISTAT for the years 2001 and 2011 only, that is when the national census was carried out. To get a value for the remaining years we assume constant rate of growth, that is a linear interpolation of the data between 2000 and 2018, as in Conti (2009), and Di Giacinto and Nuzzo (2006). Finally, we follow Bils and Klenow (2000) to construct the human capital stock of the resident population (H) in province *i*. This is proportional to an exponential function of the schooling and work experience:

$$H_i = \exp^{(\gamma_i S_i + \psi_{1,i} T_1 + \psi_{2,i} T_i^2)}$$

10) Migration flows

In recent years there has been an increase in internal migration flows of human capital, mostly from the Southern to the Northern-Centre regions (Faggian et al., 2017). On average these migrants have higher educational attainment and work experience than the population of origin, so migration flows had profound implications for the economy of the local territories (Piras, 2013). Entrepreneurial activities can also take advantages of migration flows, especially if these are of qualified human capital, because it is another way of increasing the pool of capable entrepreneurs. Moreover, migration flows, by providing a mix of people and ideas, help to strengthen entrepreneurial vitality (Florida et al., 2008). We measure "long-distance" migration flows, i.e., the difference between immigrants and emigrants to provinces outside the same NUTS-2 region, over resident population. Data on inter-provincial migration flows are taken from ISTAT, Migratory movements of resident population, registrations and cancellations to the registry office (*Movimento Migratorio della Popolazione Residente. Iscrizioni e Cancellazioni Anagrafiche*).

[21] The parameters(γ, ψ_1, ψ_2) are from Ciccone et al. (2006), who estimated an earning function using the Survey of Household Income and Wealth conducted by the Bank of Italy. As common in the literature, we assume these parameter to be time invariant and provincial specific (Aiello and Scoppa 2000, Di Giacinto and Nuzzo 2006, and Conti 2009).

[22] In the Italian school system, the number of years required to complete the primary, the lower secondary, and the upper secondary education is 5, 8, and 13 respectively. Finally, although there are different degrees and academic qualifications, in this paper we assume that to obtain a university degree the number of years required is 18.

11) Population structure

Another element that may foster an entrepreneurial ecosystem is the presence of an innovative and entrepreneurship culture. The literature suggests that young people have higher motivation and attitudes for entrepreneurship than older people (Minola, T. Criaco, G. Cassia 2014). However, data on the age of entrepreneurs are not available from ISTAT. Therefore, we use the share of young population, i.e., those aged within the 24–39 bracket, over total population, to proxy how the entrepreneurship culture is spread at provincial level. Data are from ISTAT.

b) Index construction and robustness.

Once the suitable variables have been selected, we define the normalization process, the weights, and the aggregation methods in order to construct the composite EE index. First of all, to ensure comparability and given that the variables have different measurement units, data have been normalized on a 0–10 scale using the min–max method, with higher values representing better performance:

$$\hat{x}_p = \left(\frac{x_p - min(x)}{max(x) - min(x)} \right) * 10$$

where x_p denotes the observed value of the variable in province p. In the second step, the 0–10 variable scores are averaged using equal weight to obtain an aggregate index of EE. For more comparability of provincial rankings, we further normalize the EE index using the min–max method so that the range of the EE index varies between 0 and 10. For robustness check we also use the standardization method, which converts the variables to a common scale with a mean of zero and standard deviation of one. Indicators with extreme values thus have a greater effect on the composite indicator. However, results do not show substantial changes. The Spearman rank correlation coefficients with our base EE index is 0.97.

We then select the weighting method using equal weights on the variables. It should be noted that no consensus exists on this issue. The selection of the weights implies that a judgement should be being made in terms of the relative importance of each indicator. Equal weights have some limits that can be mainly associated to its simplicity. However, equal weight is the most common scheme used in the development of composite indicators. This is especially the case when there are no statistical or empirical grounds for choosing different weights (Nardo et al., 2005). Moreover, equal weights might be a valid approach compared to other alternative weighting schemes that are based on statistical methods, such as the principal component analysis, or on subjective judgement, where weights are assigned by the researchers or by a participatory methods that involve various stakeholders, experts, citizens or politicians (see Nardo et al., 2005). Indeed, the former has some limitations especially when the variables are not highly correlated, like for some variables in our cases, while the latter requires the involvement of experts and stakeholders and the collection of their opinion on the importance of each variable. For these reasons we use equal weights to aggregate our variables.

We then proceed with the aggregation of the variables. To this purpose we choose the arithmetic mean, which is the most used aggregation method. However, we perform some robustness check by calculating the index with the geometric mean, which rewards those provinces with higher scores. Results show that changes in provincial rankings across weighting schemes are very small. The Spearman rank correlation coefficient with our base EE index is 0.98.

We perform some robustness check on the final EE index. More specifically we focus on the dimensionality issue, that is we assess to what extent the final EE index is sensitive to the inclusion/exclusion of a single variable (Nardo et al., 2005). To this purpose, given that our EE index derived from the aggregation of k dimensions ($k=11$), we construct k EE index by excluding one dimension at a time. The analysis shows that in all cases the Spearman's rank correlation coefficient with the original EE index is above 0.95, suggesting that the EE index is well-balanced in all dimensions.

Finally, we compare our EE index with a previous index calculated at the Italian NUTS-1 level, i.e., the REDI (Regional Entrepreneurial Development Index) from the national GEI (Global Entrepreneurship Index).[23] For the Italian administrative classification NUTS-1 corresponds to 4 geographical units: North-West, North-East, Centre, and South and Islands. When we aggregate our index at this spatial scale level the correlation coefficient with the REDI index calculated in the year 2007 is very high, 0.93. Also, the Spearman rank correlation coefficient is high, 0.80. This results suggests that our EE measure is well-defined.

Table 7 shows the values of the EE index for the year 2000 and 2018 at provincial level.

3) Data and descriptive statistics for model in Eq. 1

4) The distribution dynamics approach

This section summarizes the distribution dynamics approach developed by Quah (1993, 1996, 1997). This approach works in two steps. First, the probability density function of the EE index is estimated by using a kernel function (Silverman, 1986). Let $x_i, \ldots, x_n$ be a sample of n independent and identically distributed observations on a random variable X. The density value $\hat{f}(x)$ at a given point x is estimated by the following kernel density estimator:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{\| x - x_i \|}{h} \right) \tag{2}$$

where $\hat{f}(x)$ the density estimation of the variable x, h is the bandwidth (smoothing parameter) of the interval around x, and $K(\bullet)$ is the smooth and symmetric kernel function integrated to unity. Finally, $\| \bullet \|$ is the Euclidean distance. The kernel estimator assigns a weight to each observation in the interval around x with the weight being inversely proportional to the distance between the observation and x. The density estimate consists of the vertical sum of frequencies at each observation. The resulting smooth curve allows us to visualize the shape of the distribution of provincial per-capita income and detect the presence of convergence clubs represented by modes.

Second, the dynamics of the distribution and intra-distributional mobility of provinces is estimated by the stochastic kernel, which maps the density of a random variable at one point in time into the density in a subsequent period, where the density functions have been calculated through kernel density estimators (Stokey et al., 1989). In other words, the stochastic kernel describes the law of motion of a sequence of distributions and it serves to retrieve the evolution of the probability distribution of a random variable along time.

[23] For more details see https://thegedi.org/regional-gedi/ and Ortega Argiles et al., (2014).

The stochastic kernel is merely the counterpart of a transition probability matrix where the number of states (the values taken by the EE index) tends to infinity. Assuming that the frequency of the distribution follows a first order Markov process, in a transition probability matrix M each element indicates the probability that a province that was in state i in period t ends up in state j at $t+1$. Therefore, given the distribution of the EE index across provinces at time $t\left(EE_t\right)$ then the evolution of EE can be described by the law of motion: $EE_{t+1} = M * EE_t$.

Similarly to the transition probability matrix, the stochastic kernel provides the likelihood of transiting from one place in the range of values of the EE index to the others in a continuous state-space setting (Durlauf & Quah, 1999). More formally, let $f_t\left(x_t\right)$ and $f_{t+s}\left(x_{t+s}\right)$ denote the distribution of provincial EE at time t and $t+s$, respectively. Then the stochastic kernel that maps the distribution $f_t\left(x_t\right)$ into $f_{t+s}\left(x_{t+s}\right)$ is defined as follows:

$$f_{t+s}\left(x_{t+s}\right) = \int g\left(x_{t+s}|x_t\right) f_t\left(x_t\right) dx \tag{3}$$

where the conditional density function $g\left(x_{t+s}|x_t\right)$ (the stochastic kernel) describes the probability of the transition to a certain state in $t+s$ given the initial state in t. In line with (Hyndman et al., 1996), the conditional density is estimated using a kernel estimator given by:

$$\hat{g}\left(x_{t+s}|x_t\right) = \frac{\hat{z}\left(x_{t+s}|x_t\right)}{\hat{f}\left(x_t\right)} \tag{4}$$

where $\hat{f}\left(x_t\right)$ is the marginal density from Eq. (1), and $\hat{z}\left(x_{t+s}|x_t\right)$ is the joint density given by:

$$\hat{z}\left(x_{t+s}|x_t\right) = \frac{1}{nhb}\sum_{i=1}^{n} K\left(\frac{\parallel x_{t+s} - x_{i,t+s} \parallel}{b}\right)\left(\frac{\parallel x_t - x_{i,t} \parallel}{h}\right) \tag{5}$$

with b and h denoting the bandwidth parameters of the interval around x and y respectively and K is the kernel function. Finally, if the law of motion of Eq. (2) is time-invariant, then we can compute the ergodic (long-run) distribution (Johnson, 2005):

$$f_\infty\left(x_{t+s}\right) = \int g\left(x_{t+s}|x_t\right) f_\infty\left(x_t\right) dx \tag{6}$$

Equation (5) describes the convergence to a steady state distribution independent of the initial distribution. Accordingly, the ergodic distribution allows to analyze the long-run tendencies of provincial EE assuming that the observed dynamics continue to hold. This long-run equilibrium distribution plays a central role in the analysis of provincial convergence. At any point of time, the distribution $f_{t+s}\left(x_{t+s}\right)$ may reflect disequilibrium due to short-run external shocks. In the long-run, however, the ergodic distribution, $f_\infty\left(x_{t+s}\right)$, remains invariant and all transitionary effects disappear. Moreover, a tendency of $f_\infty\left(x_{t+s}\right)$ towards a unique mode, means distributional convergence. On the other hand, a divergent tendency of $f_\infty\left(x_{t+s}\right)$ towards bimodality (or multimodality), means polarization (convergence clubs) (Galor, 1996).

Finally, the stochastic kernel can also be used to describe movements between two distributions, i.e. an actual unconditional distribution and a conditional distribution (Quah, 1996). The latter is the distribution that prevails when other variables affect future values of the EE index. In this case, when the mass of probability lies along the main diagonal the conditioning factor is not responsible for the variation in the actual

distribution. In contrast, when a factor is responsible for most of the dispersion, conditioning out its effect will result in a kernel parallel to the unconditional distribution (so that provinces share similar values of the EE index when the effect of the factor is removed, regardless of their position in the original distribution).

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